

Exploring Practical Applications of Open Set and Partial Domain Adaptation in HAR

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Abkürzungsverzeichnis

UR	Universität Rostock.
HAR	Human Activity Recognition
DA	Domain Adaptation
UDA	Unsupervised Domain Adaptation
OSDA	Open-Set Domain Adaptation
PDA	Partial Domain Adaptation
UniDA	Universal Domain Adaptation
OS*	Open-Set Known-Class Accuracy
UNK	Unknown-class Accuracy
UniDABench	Universal Domain Adaptation Benchmark
CNN	Convolutional Neural Network
FNO	Fourier Neural Operator
FFT	Fast Fourier Transform
rFFT	Real Fast Fourier Transform
GRL	Gradient Reversal Layer
IMU	Inertial Measurement Unit
KL	Kullback–Leibler Divergence
MLP	Multi-Layer Perceptron
MMD	Maximum Mean Discrepancy
OT	Optimal Transport
t-SNE	t-distributed Stochastic Neighbor Embedding

Zusammenfassung

Universal Domain Adaptation (UniDA) generalises Open-Set (OSDA) and Partial (PDA) adaptation by allowing both source-private and target-private classes simultaneously. While extensively studied for images, UniDA remains under-explored for time-series Human Activity Recognition (HAR), and existing work typically evaluates each method only on its own native label-space scenario, leaving open how OSDA-, PDA-, and UniDA-designed methods compare against each other under identical conditions.

This thesis presents a controlled cross-paradigm benchmark of state-of-the-art domain adaptation methods on three HAR datasets across all four scenarios: closed-set, OSDA, PDA and UniDA. We re-implement OSBP, TSFA, SPADA and PDAAN for time-series and integrate them with seven UniDA methods and eighteen closed-set baselines under a shared training, evaluation and hyper-parameter protocol. To probe robustness, a hardest-first strategy grows the private-class count, ranked by FNO mean cosine distance to the shared knowns.

We find that the principal axis separating successful from failing methods is not the universality of their label-space assumption but the rigidity of their rejection mechanism: fixed-threshold methods (DANCE, PPOT) collapse under cross-dataset confidence shift, while adaptive or threshold-free methods (UniJDOT, UDA, OSBP) remain competitive with, and at high private-class counts surpass, scenario-native methods. UniJDOT is the only method that degrades gracefully across all four phases.

1 Introduction

Deep Learning has seen outstanding advancements in the last decade. However, deep learning faces many limitations. The biggest of which is the cost, and effort required to obtain labeled data for the supervised learning [17]. Another limitation is that deep learning relies on the assumption that the training (source), and testing (target) data are drawn from the same feature distribution. Unfortunately, this assumption doesn't hold for many real-world situations. This shift between the domains leads to a degradation in the model's ability to generalize. For Human Activity Recognition datasets, the domain shifts can be as follows:

- **Cross-Subject:** Different subjects perform the same activity (e.g., "walking") in unique ways, resulting in different sensor data patterns
- **Cross-Device:** Data from a sensor on the wrist has a different distribution than data from a sensor on the chest or ankle.
- **Cross-Environment:** Data from a controlled "lab"(source) is cleaner and more uniform than data from an in-the-wild"(target) setting.

Combination of the above factors: a domain shift could be Cross-Dataset, which means Cross subject, device, and environment at the same time. Domain Adaptation (DA) seeks to fix these problems by solving the domain discrepancy between labeled and unlabeled domains. DA means transferring knowledge from the source domain with labeled data to the target domain which has limited or no labels [18]. When the target domain has a limited number of labels, it is called "semi-supervised domain adaptation", and when it is completely unlabeled, it is called "unsupervised domain adaptation" which is the type we will be focusing on. This approach is particularly helpful for leveraging available large datasets without the expensive process of human labeling. Many studies approach the DA issue assuming that the label spaces of the source and target data are identical, meaning they have the same classes (labels). Such an approach is called **closed-set domain adaptation (CSDA)**. This approach can be impractical in many realistic situations. Especially when the two domains are of different datasets; it is unlikely that they share the exact same label space. When the target domain has more classes than the source, it is known as open set. An example of this is in the field of human activity recognition would be the source domain's label space having sitting, walking, running and the target domain's label space having sitting, walking, running, biking. A closed-set domain adaptation method would not perform well due to negative transfer. **Negative Transfer** occurs when domain adaptation actually makes the model's performance worse than if we hadn't used adaptation [16]. And in the context of an open set problem, it means the algorithm tried to forcefully align an "Unknown" or "Private" target sample with a "Known" or "Shared" source class. This means we must use algorithms which assume that the label space of both domains is not shared and try to separate the "Unknown" from the "Known" samples. Such algorithms belong to the categories of:

- **Open-Set Domain Adaptation (OSDA):** The source label space is a subset of the target label space ($Y_s \subset Y_t$). For example, a model trained on {"walking", "sitting"} is deployed in a real-world setting where it encounters {"walking", "sitting", "biking"}. The challenge is to correctly classify the shared activities while safely rejecting "biking" as an "other" or "novel" class.
- **Partial Domain Adaptation (PDA):** The target label space is a subset of the source label space ($Y_t \subset Y_s$). For example, a large source model trained on 10 activities is adapted for a target application that only cares about {"walking", "running"}. The challenge is to adapt only the relevant classes and avoid confusion from the irrelevant "source-only" classes.
- **Universal Domain Adaptation (UniDA):** This is the most generalized and challenging setting, where no prior knowledge regarding the relationship between the source and target label spaces is assumed. The domains share some common classes, but both the source and target domains may possess their own private, unshared classes ($Y_s \cap Y_t \neq \emptyset$, but neither is necessarily a subset of the other). For example, a source domain trained on {"walking", "sitting", "running"} is adapted to a target domain containing {"walking", "sitting", "biking"}. The challenge here is twofold: the model must automatically discover the shared label space (walking, sitting) to perform alignment, while simultaneously identifying and rejecting the private, "unknown" classes in both domains (running from the source, "biking" from the target) to prevent negative transfer.

In our research we will focus on analyzing the state-of-the-art domain adaptation methods in Open Set, Partial, and Universal scenarios. The type of data we will run our experiments on is time-series data, specifically Human Activity Recognition (HAR).

Our contributions to this specific research area can be summarized in the following points:

- **OSBP for HAR:** Re-implemented OSBP [45] for time-series data by adapting its adversarial unknown-rejection objective and gradient-reversal head to a 1D sequence encoder.
- **TSFA implementation:** Full from-paper re-implementation of TSFA[53], including its feature augmentation strategy, integrated into the AdaTime training pipeline.
- **PDAAN and SPADA:** [60][32] from-paper re-implementation of both methods in PyTorch and unified them for fair comparison under identical training and tuning settings.
- **Cross-dataset evaluation:** Benchmark of all methods across six domain pairs (RealWorld, PAMAP2, MHEALTH) under closed-set, OSDA, PDA, and UniDA settings with fixed seeds and shared label space.
- **Strategy of increasing unknown classes:** Controlled increase of task difficulty by progressively adding target-, source-, or both-side private classes ranked by FNO feature distance to shared classes.
- **Cross-paradigm comparison:** Systematic comparison of OSDA-native, PDA-native, and UniDA methods to isolate the effect of different adaptation assumptions.

2 Literature Review

This chapter reviews the foundations necessary to situate open-set, partial, and universal domain adaptation for cross-dataset human activity recognition. We begin with the HAR problem and cross-dataset distribution shifts, then establish the theoretical and methodological foundations of domain adaptation. We progressively narrow from general DA to time-series-specific methods, label-space mismatch formalizations, and the specific open-set, partial, and universal DA methods most relevant to this work. We utilized the help of multiple surveys and reviews for many of the references and theoretical overview.

2.1 Human Activity Recognition and the Cross-Dataset Problem

Human Activity Recognition (HAR) uses data collected from wearable inertial measurement units (IMUs), typically accelerometers and gyroscopes embedded in smartphones or dedicated sensor platforms, to classify physical activities such as walking, sitting, running, or climbing stairs [5, 25]. A HAR sample is a multivariate time series $\mathbf{x} \in \mathbb{R}^{T \times d}$, where T denotes the number of time steps in a fixed-length sliding window and d is the number of sensor channels [47]. For instance, a tri-axial accelerometer and tri-axial gyroscope yield $d = 6$ channels, and a window of 2.56 seconds at 50 Hz produces $T = 128$ samples.

Several publicly available datasets have become standard benchmarks for HAR research. PAMAP2 [41] records data from 9 subjects wearing wireless IMUs at three body locations (wrist, chest, ankle) at 100 Hz across 18 activities. RealWorld HAR [50] captures 7 activities from 15 participants with sensors at seven body positions at 50 Hz. HHAR [48] collects 6 activities from 9 users instrumented with 8 smartphones and 4 smartwatches, with sampling rates varying between 50 and 200 Hz. UCI-HAR [1] provides smartphone accelerometer and gyroscope data from 30 volunteers performing 6 activities. MHEALTH [2] records 12 activities from 10 subjects using sensors at the chest, wrist, and ankle. Opportunity [42] includes data from 4 participants performing 5 activities with accelerometers at 19 body positions. WISDM [28] captures 3-axis accelerometer measurements from 30 participants at 20 Hz across six everyday activities.

The domain gaps in HAR arise from multiple sources [47, 16]. Cross-subject variability means that even identical activities produce different sensor signatures across individuals due to differences in gait, body composition, and movement habits. Cross-device shifts emerge because different sensor hardware produce signals with distinct noise profiles and dynamic ranges [48]. Cross-position shifts occur when sensors are placed on different body parts, since the same activity generates qualitatively different acceleration patterns at the wrist versus the ankle [47]. Cross-environment shifts arise when data collected in controlled laboratory settings differs systematically from data gathered in naturalistic, “in-the-wild” conditions.

The most challenging scenario, and the one addressed in this thesis, is cross-dataset adaptation, which compounds all of the above: different subjects, different sensor hardwa-

re, different body placements, and different collection environments simultaneously. Kfari and Lüdtké [27] demonstrated that this compound shift causes severe performance degradation, with models trained on one HAR dataset (e.g., PAMAP2) failing to generalize to another (e.g., RealWorld HAR) without adaptation, even when the two datasets share activity classes. Their analysis also revealed that activities with similar kinematic profiles, such as sitting versus lying, or walking versus Nordic walking, are particularly susceptible to confusion under domain shift, underscoring the difficulty of learning transferable representations from sensor time series.

2.2 Domain Adaptation Foundations

Domain adaptation (DA) formalizes the problem of learning under distribution shift between a labeled source domain $\mathcal{D}_s = \{(\mathbf{x}_i^s, y_i^s)\}_{i=1}^{n_s}$ and an unlabeled target domain $\mathcal{D}_t = \{\mathbf{x}_j^t\}_{j=1}^{n_t}$, where the source and target are drawn from joint distributions P_s and P_t with $P_s \neq P_t$ [3]. The objective is to learn a hypothesis $h : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes the target risk $\epsilon_t(h) = \mathbb{E}_{(X,Y) \sim P_t}[\ell(h(X), Y)]$ despite having no labeled target data.

The theoretical foundation for DA rests on the generalization bound of Ben-David u. a. [3]:

$$\epsilon_t(h) \leq \epsilon_s(h) + \frac{1}{2}d_{\mathcal{H}\Delta\mathcal{H}}(P_s, P_t) + \lambda^* \quad (2.1)$$

where $\epsilon_s(h)$ is the source risk, $d_{\mathcal{H}\Delta\mathcal{H}}(P_s, P_t)$ measures the divergence between source and target distributions under a symmetric difference hypothesis class, and λ^* is the error of the ideal joint classifier. This bound motivates three adaptation principles: minimizing source error through supervised training, reducing domain discrepancy through feature alignment, and controlling the joint error through appropriate hypothesis class selection [16].

Following the categorization in He u. a. [20], DA methods can be organized into three families.

Adversarial training methods introduce a domain discriminator that is optimized to distinguish source and target domains, while a deep classification model learns transferable features indistinguishable by the discriminator [17, 21, 52, 34]. Domain-Adversarial Neural Networks (DANN) [17] implemented this principle using a gradient reversal layer, which negates gradients flowing from the discriminator to the feature extractor during backpropagation, enabling end-to-end training. The adversarial loss takes the form:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{\mathbf{x}_i^s \in \mathcal{D}_s} \log[G_d(\mathbf{z}_i^s)] + \mathbb{E}_{\mathbf{x}_i^t \in \mathcal{D}_t} \log[1 - G_d(\mathbf{z}_i^t)] \quad (2.2)$$

The feature extractor minimizes $\mathcal{L}_c + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}}$ while the discriminator maximizes $\mathcal{L}_{\text{adv}}$, resulting in domain-invariant features through this minimax optimization [47].

Statistical divergence methods aim to extract domain-invariant features by minimizing domain discrepancy in a latent feature space. Widely used measures include Maximum Mean Discrepancy (MMD) [33, 43], correlation alignment (CORAL) [49], and optimal transport (OT) distances [10, 40]. MMD compares distributions through their mean embeddings in a reproducing kernel Hilbert space:

$$\mathcal{L}_{\text{MMD}} = \left\| \mathbb{E}_{P_s}[\phi(\mathbf{z}^s)] - \mathbb{E}_{P_t}[\phi(\mathbf{z}^t)] \right\|_{\mathcal{H}_k}^2 \quad (2.3)$$

OT computes the minimum-cost plan for moving probability mass from source to target,

yielding the Wasserstein distance:

$$W_c(P_s, P_t) = \inf_{\gamma \in \Pi(P_s, P_t)} \int_{\mathcal{X} \times \mathcal{X}} c(x_s, x_t) d\gamma(x_s, x_t) \quad (2.4)$$

where $c(x_s, x_t)$ is the transport cost and $\Pi(P_s, P_t)$ is the set of all couplings with marginals P_s and P_t . Sinkhorn divergence [11] enables efficient computation through entropic regularization, interpolating between the pure OT loss and MMD [15]. A practical advantage of OT over MMD is its gradient behavior: MMD with an RBF kernel produces vanishing gradients when source and target distributions are far apart, while OT maintains strong gradient signal regardless of the separation between distributions [20, 15]. Additionally, the optimal transport plan γ^* provides soft assignments between source and target samples, which becomes useful for open-set and universal adaptation where unmatched mass can signal unknown classes.

Self-supervision methods incorporate auxiliary self-supervised training tasks to learn domain-invariant features through pretext objectives that do not require target labels [26]. These include contrastive learning objectives, augmentation-based pretext tasks, and reconstruction-based methods that achieve alignment by jointly reconstructing source and target data [19, 24]. Self-supervised approaches are often combined with statistical divergence or adversarial methods to improve feature transferability.

2.3 Domain Adaptation for Time Series

While domain adaptation originated in computer vision, time-series data introduce additional challenges. Time-series samples $\mathbf{x} \in \mathbb{R}^{T \times d}$ exhibit domain shifts not only in feature distributions but also in temporal dynamics and spectral representations. Differences in sampling rate, sensor characteristics, and recording conditions alter both the marginal feature distribution $P(X)$ and the conditional mapping $P(Y|X)$, producing discrepancies in sequence patterns, frequency characteristics, and inter-channel dependencies [22, 16].

2.3.1 Benchmarking and Evaluation

The ADATIME benchmark [39] established the first standardized evaluation suite for unsupervised DA on temporal data, providing a unified experimental framework with consistent preprocessing, fixed source-target splits, and comparable evaluation across HAR, physiological, and industrial datasets. ADATIME compared representative methods from adversarial, discrepancy-based, and contrastive families under controlled conditions. Its codebase, built around a 1D-CNN backbone (which was shown to consistently outperform more complex architectures such as ResNet-18 and TCN), uses a fixed backbone across all methods to ensure that differences in performance can be attributed to the adaptation algorithm. The ADATIME framework has become the standard implementation baseline for subsequent time-series DA research [20, 37].

Building on the ADATIME codebase, the UniDABench benchmark [36] extended standardized evaluation to the universal DA setting on time series. UniDABench introduced a protocol for artificially creating UniDA tasks from existing HAR, HHAR, and Sleep-EDF datasets by removing classes to simulate label-space mismatch. This benchmark tested six UniDA methods across four backbone architectures (CNN, FNO, TSLANet, S3), revealing that CNN and FNO consistently outperform newer time-series architectures for

domain adaptation, and that UniJDOT is the most robust method across all backbones.

2.3.2 Closed-Set Time-Series DA Methods

Several methods have been developed specifically for closed-set time-series DA. CoDATS [56] extended DANN to time series with a convolutional backbone and weak supervision for multi-source aggregation. AdvSKM [31] aligned domain feature spectra through a hybrid spectral kernel matching technique. CALDA [55] combined contrastive and adversarial learning for multi-source time-series DA. SASA [6] captured inter-class temporal dependencies through sparse associative structure alignment. CLUDA [37] proposed a contrastive learning framework that extracts contextual representations preserving label information across domains. CoTMix [13] generalized contrastive temporal adaptation through self-supervised temporal mixing.

Kfari und Lütke [27] applied the SelfHAR [51] teacher-student self-training pipeline to cross-dataset HAR transfer. In SelfHAR, a teacher network generates pseudo-labels for unlabeled target data, a confidence-based selection mechanism retains high-confidence predictions, and a student network trains on these pseudo-labeled samples using a multi-task objective combining activity classification with augmentation discrimination. Kfari und Lütke [27] found that this self-training pipeline implicitly performs domain adaptation even without explicit alignment objectives, achieving F1 scores around 0.8 for cross-dataset transfer between PAMAP2, RealWorld HAR, and MHEALTH.

2.3.3 Time-Frequency Features for Domain Adaptation

An important finding in recent time-series DA research is that frequency features are more domain-invariant than temporal features. He u. a. [20] demonstrated this empirically: for the same activity performed by different individuals, the Fourier representations show substantially less cross-domain divergence than the raw time-domain signals.

RAINCOAT [20] exploited this insight by employing a dual encoder: a temporal CNN for time-domain features and a frequency encoder inspired by the Fourier Neural Operator (FNO) [30] for frequency-domain features. For domain alignment, RAINCOAT uses Sinkhorn divergence, providing theoretical and empirical justification that Sinkhorn maintains stronger gradients when aligning frequency features whose amplitude distributions may have disjoint supports across domains.

This time-frequency encoding principle has been independently validated by subsequent work. TSFA [53] incorporated a learnable weighting mechanism over FFT magnitude modes to adaptively select transferable frequency components. The UniDABench benchmark [36] further demonstrated that FNO-based backbones consistently outperform newer architectures for domain adaptation. These converging findings establish time-frequency encoding as a general design principle for time-series DA.

2.4 Label-Space Mismatch in Domain Adaptation

The methods reviewed thus far assume that the source and target domains share identical label spaces ($\mathcal{Y}_s = \mathcal{Y}_t$). This assumption rarely holds in practice. Naively applying closed-set DA methods under label-space mismatch leads to negative transfer: the algorithm forcibly aligns unknown target samples with known source classes, degrading performance on both.

Four canonical label-space configurations exist [18, 16, 59]:

Closed-set DA ($\mathcal{Y}_s = \mathcal{Y}_t$): Source and target share identical label spaces. The Ben-David bound (Equation 2.1) applies directly.

Partial DA (PDA) ($\mathcal{Y}_t \subset \mathcal{Y}_s$): The target label space is a strict subset of the source. Selective reweighting or pruning of source-only classes ($\mathcal{Y}_s \setminus \mathcal{Y}_t$) mitigates negative transfer.

Open-set DA (OSDA) ($\mathcal{Y}_s \subset \mathcal{Y}_t$): The target domain contains unknown classes absent from the source. The OSDA problem was first introduced by Saito u. a. [45], but a formal theoretical upper bound for the target domain risk was not established until the work of Fang u. a. [14]. Both theoretical developments are comprehensively reviewed by Ghaffari u. a. [18].

Universal DA (UniDA) ($\mathcal{Y}_s \cap \mathcal{Y}_t \neq \emptyset$, with private classes on both sides): The most general setting, subsumes all others. The risk separates into shared and unknown components [16]:

For HAR, the label-space mismatch problem is particularly acute in cross-dataset scenarios. Adapting from PAMAP2 (18 activities) to RealWorld HAR (8 activities) involves both source-private and target-private classes, placing cross-dataset HAR squarely in the universal DA setting.

2.4.1 Evaluation Metrics

Standard accuracy is insufficient for evaluating open-set and universal DA [18]. OS* measures accuracy only on known-class target samples. UNK measures the fraction of truly unknown samples correctly identified. HOS computes the harmonic mean of OS* and UNK. H-score, used in the universal setting, computes the harmonic mean of accuracy on common classes (CA_c) and accuracy on private classes (CA_u): $\text{H-score} = (2 \cdot CA_c \cdot CA_u) / (CA_c + CA_u)$ [20].

2.5 Open-Set, Partial, and Universal DA Methods

2.5.1 Open-Set Domain Adaptation

Ghaffari u. a. [18] categorize OSDA methods for computer vision into adversarial methods (e.g., OSBP [45], UADAL [23]), metric-based methods, informative feature extraction methods (e.g., ROS [4]), and hybrid methods (e.g., OVANet [44], GLC++ [38], COSDA [54]). All were developed on image benchmarks.

For time-series data, TSFA [53] proposes a two-stage multigranularity alignment framework for OSDA: a global stage minimizing similarity distribution entropy (SDE) on a hypersphere, followed by a local stage using pseudo-labels from an optimal similarity assignment matrix (OSAM) formulated as a global OT problem. TSFA uses adaptive decision boundaries from the OSAM transport plan rather than fixed thresholds, and achieves H-scores of 92.42%, 88.61%, and 87.12% on HAR, CWRU, and BEAR respectively, outperforming RAINCOAT and OVANet in the OSDA setting.

2.5.2 Partial Domain Adaptation

In computer vision, PADA [7] introduced class-level weighting with adversarial learning, and ETN [8] learned an explicit transferability network.

For time-series fault diagnosis, PDAAN [60] addresses bearing fault diagnosis under class-space asymmetry using class-level weights, weighted complementary entropy, and domain enhancement. SPADA [32] uses a weighted domain discriminator based on source-target similarity estimation. Both address partial DA exclusively for mechanical fault diagnosis and have not been applied to HAR.

2.5.3 Universal Domain Adaptation

UAN [59] introduced transferability scoring to selectively align shared classes. DANCE [46] extended this with self-supervised neighborhood clustering. UniOT [9] used unbalanced OT with adaptive filling for threshold-free common/private detection.

RAINCOAT [20] is the first model for both closed-set and universal DA on time series. Beyond time-frequency encoding (Section 2.3.3), it introduces an “align-then-correct” strategy: after alignment, the encoder is retrained on target data via reconstruction, and the differential movement of features before and after correction separates known from unknown samples through bimodality detection. RAINCOAT achieved up to 16.33% H-score improvement over prior methods.

UniJDOT [35] extends DeepJDOT [12] to the universal setting for time series. It addresses classifier overconfidence on unknown samples [29] by constructing a joint decision space combining classifier logits with distance-based probabilities from classwise memory banks, and uses auto-thresholding (Yen’s method) to determine the common/unknown split per batch. Common samples are aligned via OT with unknown-aware cost rewriting. UniJDOT achieved the highest average H-scores across HAR (62%), HHAR (53%), and EDF (51%) in the UniDABench evaluation [36].

3 Foundations

All experiments are conducted on three publicly available human-activity recognition (HAR) datasets: **RealWorld**, **PAMAP2**, and **mHealth**. These were selected for the variety of their sensor placements, rich activity selection, and data quality.

Across these experiments, we evaluate every ordered source–target pair drawn from the three datasets, resulting in six evaluation scenarios:

- RealWorld $\rightarrow$ PAMAP2, RealWorld $\rightarrow$ mHealth
- PAMAP2 $\rightarrow$ RealWorld, PAMAP2 $\rightarrow$ mHealth
- mHealth $\rightarrow$ RealWorld, mHealth $\rightarrow$ PAMAP2

4 activities common to all three datasets, *lying*, *sitting*, *walking*, and *running*, are designated as the **shared known classes** and form the closed-label space. Every method is trained from scratch using a shared Fourier Neural Operator (FNO) backbone and is repeated across five independent random seeds and 20 epochs; results are reported as mean $\pm$ standard deviation. We ran all the experiments with the help of an NVIDIA RTX 5000.

Evaluation Metrics

We report Macro F1-score for closed-set and PDA experiments, and H-score for OSDA and UniDA experiments. For completeness, we also report OS* and UNK.

Macro F1 is defined as:

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \text{F1}_k \quad (3.1)$$

where K is the number of classes.

For OSDA and UniDA, OS* measures accuracy on shared classes, while UNK measures accuracy on target-private (unknown) samples. The primary metric is the H-score:

$$H = \frac{2 \cdot \text{OS}^* \cdot \text{UNK}}{\text{OS}^* + \text{UNK}} \quad (3.2)$$

which balances shared-class recognition and unknown-class rejection via the harmonic mean.

3.1 Datasets

- **RealWorld HAR Dataset**[50]
 - **Participants:** 15 subjects.
 - **Gender Distribution:** 8 males and 7 females.

- **Body Positions:** 7 positions: chest, forearm, head, shin, thigh, upper arm, and waist.
- **Sensors:** Smartphones and smartwatches providing data from accelerometers, gyroscopes, magnetometers, and GPS.
- **Frequency:** 50 Hz.
- **Activities:** 8 activities: walking, jogging, sitting, standing, lying down, climbing stairs up, climbing stairs down, and jumping.
- **PAMAP2 (Physical Activity Monitoring for Aging People) Dataset**[41]
 - **Participants:** 9 subjects.
 - **Gender Distribution:** 8 males and 1 female.
 - **Body Positions:** 3 positions: dominant wrist, ankle, and chest.
 - **Sensors:** 3 Inertial Measurement Units (IMUs) containing two 3-axis accelerometers, a 3-axis gyroscope, and a 3-axis magnetometer. A heart rate (HR) monitor is also included.
 - **Frequency:** 100 Hz for the IMU sensors and approximately 9 Hz for the heart rate monitor.
 - **Activities:** 18 total activities. This includes 12 protocol activities (lying, sitting, standing, walking, running, cycling, Nordic walking, ascending stairs, descending stairs, vacuum cleaning, ironing, and rope jumping) and 6 optional activities (watching TV, computer work, car driving, folding laundry, house cleaning, and playing soccer). The optional activities are never used or reported in the actual dataset.
- **MHEALTH (Mobile HEALTH) Dataset**[2]
 - **Participants:** 10 subjects.
 - **Gender Distribution:** Not specified.
 - **Body Positions:** 3 positions: chest, right wrist, and left ankle.
 - **Sensors:** 3 Shimmer2 wearable sensors (providing 3-axis acceleration, rate of turn, and magnetic field data), along with a 2-lead ECG sensor attached to the chest.
 - **Frequency:** 50 Hz.
 - **Activities:** 12 activities: standing still, sitting and relaxing, lying down, walking, climbing stairs, waist bends forward, frontal elevation of arms, knees bending (crouching), cycling, jogging, running, and jump front & back.

3.2 Preprocessing

- **Resampling:** All datasets are downsampled to a shared 50 Hz to standardize temporal resolution across sources (original rates are 50 Hz for RealWorld and mHealth, 100 Hz for PAMAP2). This ensures the window length corresponds to the same time duration across datasets.

- **Sensors and axes:** We retain three full inertial sensors—**accelerometer**, **gyroscope**, and **magnetometer**—each with **x**, **y**, **z** axes.
- **Body positions:** We choose the **two closest shared body positions** across datasets (forearm/hand and shin/ankle equivalents) to maximize cross-dataset compatibility.
- **Feature count:** With $2 \text{ positions} \times 3 \text{ sensors} \times 3 \text{ axes}$, each time step has 18 features.
- **Windowing logic:** Signals are segmented into fixed-length windows of 150 samples with a 50% overlap (shift of 75 samples). At 50 Hz, each window spans exactly 3 seconds.
- **Label aggregation:** Each window receives a single class label by taking the most frequent label (majority vote) among the 150 samples inside the window.
- **Normalization:** For each sensor channel independently, compute the dataset-level mean μ_c and standard deviation σ_c over all windows (and all time points within those windows). Each sample is standardized as $x' = \frac{x - \mu_c}{\sigma_c}$. This yields zero-mean, unit-variance inputs per channel before training and prevents any single sensor or axis from dominating due to scale differences.

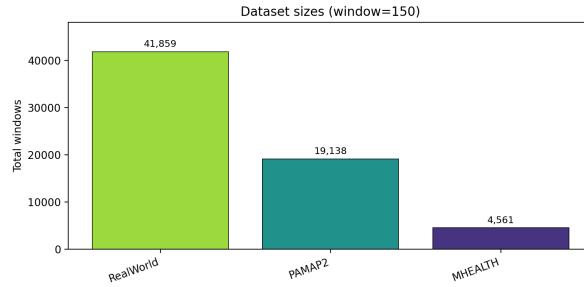


Abbildung 3.1: Size of datasets in windows

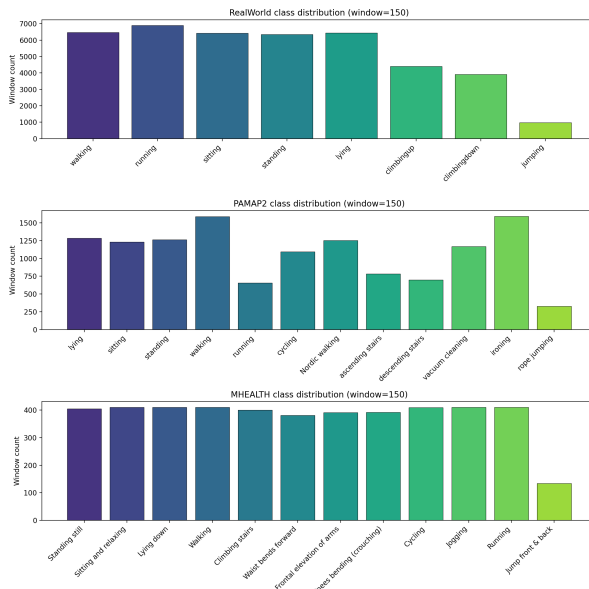


Abbildung 3.2: Class Distributions of the Datasets

3.3 Hyperparameter Search

3.3.1 Hyperparameter Optimization Strategy

For the hyperparameter optimization strategy, we had a choice between conducting a cross-dataset sweep or a cross-user sweep. While a cross-dataset sweep would have yielded hyperparameters more finely tuned to inter-dataset distribution shifts, it risks overfitting the hyperparameters to the specific dataset pair used for the sweep, which would necessitate discarding the dataset pair.

We instead opted for a cross-user sweep (specifically, Male $\rightarrow$ Female) within the RealWorld dataset. While using a dataset that will eventually be featured in the final cross-dataset training scenarios carries a slight risk of overfitting, this risk is substantially lower in the context of domain adaptation. Because the final evaluation tests adaptation across entirely different domains (e.g., varying sensor hardware and placements across datasets), overfitting to the specific physiological variance of the RealWorld users does not meaningfully compromise the evaluation of the algorithm’s cross-dataset transferability.

Sweep Configuration

The hyperparameter optimization was conducted using the following setup:

- **Framework:** 15 Bayesian Optuna trials.
- **Range:** The hyperparameter ranges to search through were taken from the codebase of UniDABench for the UniDA methods, and for the OSDA, and PDA methods we used the hyperparameter ranges of their papers. The algorithms which we didn’t find a range for, we approximated. The ranges can be found in Table 3.1.
- **Execution:** Each trial executes 3 random seeds and averages the resulting metrics to ensure stability.
- **Objectives:** The optimization objective maximizes the average **H-score** for Universal Domain Adaptation (UniDA) tasks, and maximizes the average **Macro F1-score** for Partial Domain Adaptation (PDA) tasks.
- **Logging:** MLflow is utilized to comprehensively record the hyperparameters, per-run metrics, and final trial averages.

RealWorld Subject Split (Male $\rightarrow$ Female)

The sweep utilizes a cross-gender distribution shift within the RealWorld dataset:

- **Source Domain (Male Subjects):** Subjects 2, 3, 4, 5, 7, 9, 10, and 14.
- **Target Domain (Female Subjects):** Subjects 1, 6, 8, 11, 12, 13, and 15.

Activity Splits by Paradigm

To properly evaluate both paradigms, the activities were partitioned to meet their respective structural assumptions:

Universal Domain Adaptation (UniDA) Split:

- **Shared Classes (4):** sitting, lying, running, walking.
- **Source-Private (2):** climbing down, jumping.
- **Target-Private (2):** standing, climbing up.

This configuration yields 6 total source-known classes. During evaluation, the target-private classes are aggressively treated as "unknown" to test the algorithm's rejection capabilities.

Partial Domain Adaptation (PDA) Split:

- **Shared Classes (4):** sitting, lying, running, walking.
- **Source-Private (2):** climbing down, jumping.
- **Target-Private (0):** None.

This configuration strictly enforces the fundamental PDA assumption: the target label space must be a strict subset of the source label space ($Y_t \subset Y_s$).

Tabelle 3.1: Hyperparameter Search Spaces for Optuna Sweeps

Category / Algorithm	Hyperparameter	Search Space / Range
General Training	batch_size	$\{32, 64\}$
	weight_decay	$\{10^{-4}, 5 \times 10^{-4}\}$
UDA	learning_rate	$\{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	src_cls_loss_wt	$[0.05, 5.0], \text{step} = 0.05$
	domain_loss_wt	$[0.05, 5.0], \text{step} = 0.05$
DANCE	learning_rate	$\{10^{-2}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	eta	$[0.05, 1.0], \text{step} = 0.05$
	margin	$[0.05, 1.0], \text{step} = 0.05$
UniOT	learning_rate	$\{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	gamma	$[0.05, 1.0], \text{step} = 0.05$
	mu	$[0.05, 0.95], \text{step} = 0.05$
	temp	$\{0.05, 0.1\}$
PPOT	learning_rate	$\{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	tau	$[0.3, 0.7], \text{step} = 0.05$
	alpha, beta	$\{0.01, 0.001\}$
	ot	$[0.05, 5.0], \text{step} = 0.05$
	thresh	$[0.3, 0.9], \text{step} = 0.05$
UniJDOT	learning_rate	$\{10^{-2}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	lamb, alpha	$[0.05, 5.0], \text{step} = 0.05$
	K	$\{5, 10, 15, 20\}$
	trg_mem_size	$\{32, 64, 128\}$
RAINCOAT	learning_rate	$\{10^{-2}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	sinkhorn_eps	$\{10^{-2}, 10^{-3}, 10^{-4}\}$
TSFA	learning_rate	$\{10^{-3}, 5 \times 10^{-4}, 10^{-4}, 5 \times 10^{-5}\}$
	lambda_sde	$[0.1, 5.0], \text{step} = 0.1$
OSBP	learning_rate	$\{5 \times 10^{-3}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	lambda_adv	$[0.1, 10.0], \text{step} = 0.1$
OVANet	learning_rate	$\{0.5, 0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001\}$
SPADA	learning_rate	$\{10^{-2}, 10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$
	domain_loss_wt	$[0.05, 2.0], \text{step} = 0.05$
PDAAN	learning_rate	$\{0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001\}$
	alpha, beta	$\{0.5, 1.0, 1.5, 2.0\}$
	weights_momentum	$\{0.5, 0.7, 0.9, 0.95, 0.99\}$

3.4 Experimental Setup

Our benchmark combines two sets of methods. The first is inherited from UniDABench [cite]: UDA, OVANet, DANCE, PPOT, UniOT, and UniJDOT — the six universal-DA baselines we adopted as-is, changing only what is needed for our six HAR pair definitions, and our scenario-driven evaluator. The second set was added by us: we implement one universal method (RAINCOAT, which is not part of UniDABench despite being a key

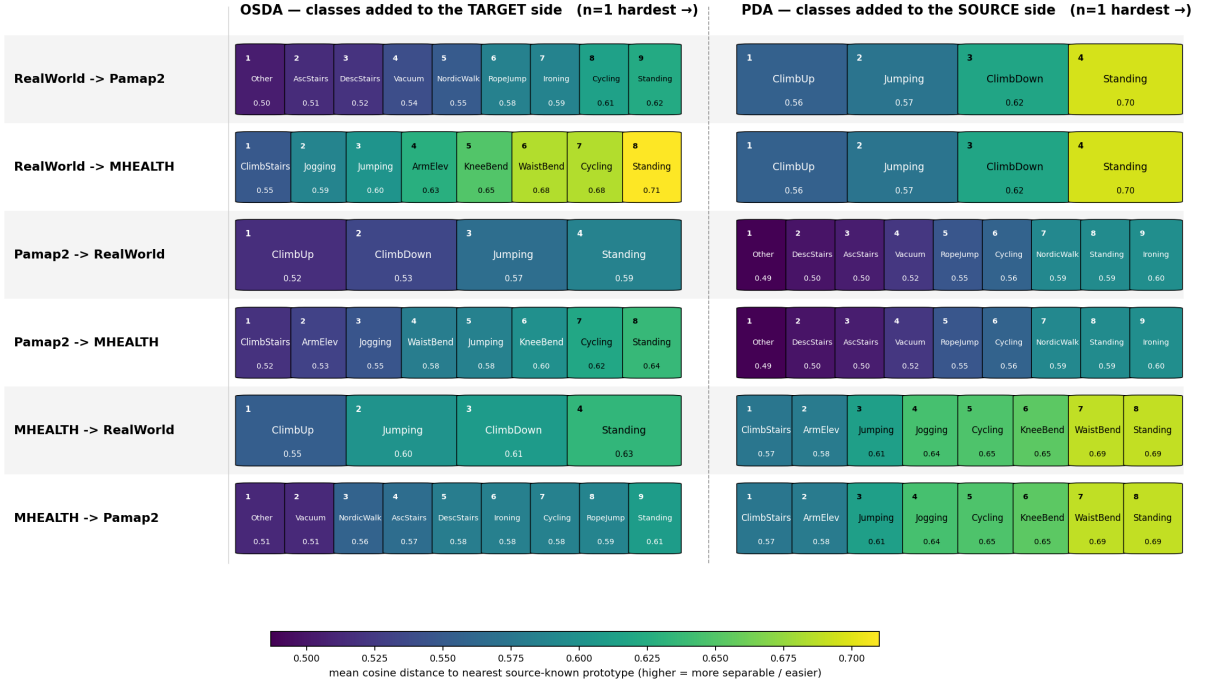


Abbildung 3.3: Difficulty ranking of all the private classes across each dataset pair, and scenario

reference for the time-series DA literature), two open-set methods (OSBP, TSFA), and two partial-DA methods (SPADA, PDAAN), so that every scenario (OSDA, PDA, UniDA) is benchmarked against at least one scenario-native method in addition to the universal baselines.

Original domain of each method

Five of the benchmark’s methods were originally designed for time series: RAINCOAT [He et al., ICML 2023] for general time-series DA with feature and label shifts, UniJDOT for time-series universal DA via joint distribution optimal transport, TSFA [TNNLS 2026] for time-series open-set DA, and SPADA [Liu et al., IEEE TII 2021] and PDAAN [Zhou et al., MST 2022] for fault-diagnosis / condition-monitoring (where the inputs are typically single-channel vibration or current signals from rotating machinery — we re-target them at multi-channel HAR). The remaining six — UDA, OVANet, DANCE, PPOT, UniOT, and OSBP [Saito et al., ECCV 2018] — were originally introduced on image-classification benchmarks (Office-31, OfficeHome, VisDA); UDA, OVANet, DANCE, PPOT, and UniOT were re-implemented for time series in UniDABench, and we adapted OSBP to time series ourselves.

Similarity Score

In our experiments we want to stress test Open Set, Partial, and Universal domain adaptation methods by increasing the private classes incrementally to the maximum to analyze how each method reacts. We chose to add the hardest or most similar private classes first. We achieve this with a simple similarity score using cosine distance and the centroid of private and shared classes feature vectors.

An easy first or random traversals are also viable options, but we wanted our performance numbers to be interpretable as strict lower bounds on the methods’ capabilities. The framework selects unknown classes based on their feature-space proximity to the four shared known classes (lying, sitting, walking, running). Unknowns located closer to the known feature clusters are considered harder to reject, as the model is inherently biased toward assigning them to the nearest known category. Conversely, distant unknowns fall into low-density regions and are easier to identify as out-of-distribution.

Similarity Score Computation

We compute the similarity score for each private class through the following steps:

- **Feature Extraction:** A feature extractor f_S (CNN+FNO backbone) is trained from scratch on the source dataset using only closed-set cross-entropy over the known classes. To mitigate single-initialization variance, this training is repeated across five random seeds, yielding embedding spaces $\{f_S^{(1)}, \dots, f_S^{(5)}\}$.
- **Centroid Extraction:** For each seed i , we compute the mean feature vector (centroid) $\mu_k^{(i)}$ for each shared known class k , as well as the centroid $\nu_p^{(i)}$ for each candidate private class p .
- **Pairwise Cosine Distance:** The distance between every private and known centroid pair is computed in feature space. Cosine distance is utilized because downstream adaptation methods rely on cosine similarity, and it ignores non-discriminative magnitude differences:

$$d^{(i)}(p, k) = 1 - \frac{\langle \nu_p^{(i)}, \mu_k^{(i)} \rangle}{\|\nu_p^{(i)}\| \|\mu_k^{(i)}\|} \quad (3.3)$$

- **Aggregation Across Seeds:** Distances are averaged across seeds at the pair level. This preserves the original metric while damping initialization noise:

$$\bar{d}(p, k) = \frac{1}{N_{\text{seeds}}} \sum_{i=1}^{N_{\text{seeds}}} d^{(i)}(p, k) \quad (3.4)$$

- **Per-Class Difficulty Score:** The overall difficulty score for a private class p is the mean of its seed-averaged distances to all known classes ($\mathcal{K}$):

$$\text{Score}(p) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \bar{d}(p, k) \quad (3.5)$$

Using the mean captures proximity to the entire known set, avoiding the outlier sensitivity that would result from simply using the single nearest class.

3.5 Experiments

Experiment 1: Closed-set scenario Across All Methods

Research Question: When the target domain contains only the four shared known classes (closed-set UDA setup), how well do open-set, partial-domain, and universal domain-

adaptation methods perform compared to dedicated state-of-the-art closed-set UDA methods? Does the specialized machinery for label-set shift impose a performance penalty when no label shift exists?

Setup: Both source and target are restricted to the four shared known classes: sitting, lying, walking, running

- **Closed-set UDA:** Baselines from the AdaTime framework.
- **OSDA:** OSBP and TSFA.
- **PDA:** SPADA and PDAAN.
- **UniDA:** UDA, OVANet, DANCE, PPOT, UniOT, UniJDOT, and RAINCOAT.

Metrics: **Macro F1-score** is the metric used. Since no target sample is private, H-score and UNK are omitted.

Experiment 2: OSDA Scenario

Research Question: As the open-set difficulty of the target grows, measured by the number of target-private classes added, how do OSDA and UniDA methods degrade? Is this degradation predictable from the feature-space similarity of the introduced privates? and do native OSDA methods such as OSBP and TSFA perform better than universal methods?

Setup: The source remains fixed at the four shared knowns. The target is augmented by introducing target-private classes in a *hardest-first* order based on the similarity ranking (n privates with the smallest mean cosine distance to source-known centroids). The range of n varies per pair based on available residual activities:

Methods: OSDA methods (OSBP, TSFA) and the UniDA family.

Metrics: **Macro F1-score**, **H-score**, OS^* (known accuracy), and **UNK** (unknowns-no accuracy).

Experiment 3: PDA Scenario

Research Question: When the source carries an increasing number of source-private distractor classes not present in the target, how well do PDA and UniDA methods suppress those distractors? and are the UniDA suppression mechanisms as effective as the dedicated PDA methods?

Setup: The target is fixed at the four shared knowns. The source is augmented using the hardest-first source-private strategy. Step n introduces the n source-privates whose centroids lie closest to the known manifold.

Methods: PDA methods (SPADA, PDAAN) and the UniDA family.

Experiment 4: UniDA Scenario

Research Question: The most general scenario, where both source-private and target-private classes are present, how do universal methods perform, and how does performance scale with the joint difficulty of privates on both sides?

Setup: Both sides grow simultaneously. At step n , the n source-privates and n target-privates from their respective rankings are introduced.

Methods: Only the UniDA family is evaluated (UDA, OVANet, DANCE, PPOT, UniOT, UniJDOT, RAINCOAT), as they are the only methods designed to handle bilateral shift.

Metrics: Macro F1-score, H-score, OS^* , and UNK.

3.6 Backbone Architecture

Following the architecture established in UniDABench [36], all domain adaptation methods in this work share a dual-branch backbone $G_\phi = G^T \oplus G^F$ that fuses a time-domain representation e_T with a frequency-domain representation e_F . Both branches consume the same raw multivariate window $x \in \mathbb{R}^{T \times C}$ ($T=150$ timesteps, $C=18$ inertial channels). Their outputs are concatenated and L_2 -normalised to form the latent representation $z = [e_F; e_T] \in \mathbb{R}^{224}$ used by all downstream classifiers and discriminators.

Time Branch (G^T)

A three-layer 1-D CNN. Each block applies `Conv1d` ($k=5$, same padding) $\rightarrow$ `BatchNorm1d` $\rightarrow$ `ReLU` $\rightarrow$ `Dropout`. The channel width progresses $18 \rightarrow 64 \rightarrow 128 \rightarrow 96$, and an `AdaptiveAvgPool1d` collapses the temporal axis to a single vector, yielding $e_T \in \mathbb{R}^{96}$.

Frequency Branch (G^F)

Human activities have characteristic periodicities, gait cadences, pedal frequencies, baseline drift during static postures, that a time-domain CNN must rediscover through learned kernels, whereas a spectral representation exposes them directly as coefficient amplitudes. Moreover, the same activity performed at slightly different speeds shifts substantially in time-domain features but remains close in the frequency domain, a property that benefits cross-dataset alignment where source and target subjects execute activities at different paces [20, 53].

We adopt the modified Fourier layer from RAINCOAT [20], which departs from the standard FNO [30] in three ways. First, a cosine smoothing $\tilde{x} = \cos(x)$ is applied before transformation to suppress spectral leakage caused by non-periodic window boundaries. Second, after computing the real FFT $v = \text{rFFT}(\tilde{x}) \in \mathbb{C}^{(T/2+1) \times C}$, only the first $M=64$ low-frequency modes are retained; this covers all activity-relevant rhythms while discarding high-frequency sensor noise that is device-specific and harmful to alignment. A learnable complex weight tensor $B \in \mathbb{C}^{C_{\text{in}} \times C_{\text{out}} \times M}$ then mixes and re-weights channels independently at each retained mode via $\tilde{v}[k] = \sum_c B[c, \cdot, k] v[k, c]$. Third, instead of inverting back to the time domain, polar coordinates are extracted: amplitude $a = |\tilde{v}|$ and phase $p = \angle \tilde{v}$. The amplitude captures how energetic each spectral mode is (encoding activity class), while the phase encodes temporal alignment within the window. The resulting $(a, p) \in \mathbb{R}^{C \times 2M}$ tensor is reduced over the channel axis by a 1×3 `Conv1d` $\rightarrow$ `BatchNorm1d` $\rightarrow$ `ReLU`, yielding $e_F \in \mathbb{R}^{128}$.

Fusion

The two streams are concatenated to produce $z \in \mathbb{R}^{96+128} = \mathbb{R}^{224}$, which is L_2 -normalised before being passed to the method-specific classifier C_θ and any auxiliary heads (domain discriminator, OVA classifier, OT prototype bank).

3.7 Algorithms

This section describes the methods evaluated in our benchmark, organized according to the label set assumption under which they were originally proposed. We begin with open set domain adaptation (OSDA), proceed to partial domain adaptation (PDA), and conclude with universal domain adaptation (UniDA), following the historical progression from early adversarial methods to more recent prototype and optimal transport based formulations.

Let the labeled source domain and unlabeled target domain be defined as

$$\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}, \quad (3.6)$$

$$\mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}. \quad (3.7)$$

Their corresponding label spaces are denoted by $\mathcal{C}_s$ and $\mathcal{C}_t$, with shared classes

$$\mathcal{C} = \mathcal{C}_s \cap \mathcal{C}_t. \quad (3.8)$$

A feature extractor G maps inputs into a latent feature space, while a classifier C predicts logits over the source label space, optionally augmented with an explicit unknown category.

3.7.1 Open Set Domain Adaptation

OSBP (Open Set Domain Adaptation by Backpropagation)

OSBP, introduced by Saito u. a. [45], is one of the earliest adversarial approaches for open set domain adaptation. The method assumes that the source domain contains only known classes,

$$\mathcal{C}_s \subset \mathcal{C}_t, \quad (3.9)$$

while target private samples must be identified and rejected without explicit unknown supervision.

The architecture consists of a feature extractor G and a classifier C whose output layer contains $K + 1$ logits, where

$$K = |\mathcal{C}_s|. \quad (3.10)$$

The additional output dimension corresponds to the unknown category.

Source samples are optimized using standard supervised cross entropy classification. For target samples, the classifier is trained to predict a fixed probability t for the unknown output, typically chosen as

$$t = 0.5. \quad (3.11)$$

The adversarial target loss is defined as

$$\mathcal{L}_{\text{adv}}(x_t) = -t \log p(y = K + 1 \mid x_t) - (1 - t) \log (1 - p(y = K + 1 \mid x_t)). \quad (3.12)$$

The classifier minimizes $\mathcal{L}_{\text{adv}}$, while the feature extractor maximizes it through adversarial optimization. Consequently, target features are encouraged either toward known source clusters or toward rejection as unknown samples. The threshold $t = 0.5$ therefore acts as a separating boundary between alignment and rejection. Optimization is performed end to end using a gradient reversal layer.

TSFA (Two Stage Feature Alignment)

TSFA, proposed by Wang u. a. [53], is a dedicated open-set domain adaptation framework specifically designed for time series classification. The method addresses two major challenges in temporal adaptation: nonstationary signals and mismatched frequency characteristics across domains.

The architecture contains three main components:

1. a time frequency feature extractor,
2. a two stage alignment mechanism,
3. and an Optimal Similarity Assignment Matrix (OSAM) for pseudo labelling.

Temporal features are extracted using one dimensional convolutional blocks followed by global average pooling. In parallel, a frequency branch computes Fourier representations and selects domain invariant spectral components using a learnable weight tensor $W \in \mathbb{R}^{C \times C \times K}$, which is applied to the top- K FFT magnitude modes via element-wise weighted summation along the frequency dimension. During training, gradients from stable frequency modes across domains accumulate constructively while gradients from domain-specific variations cancel out, enabling W to adaptively emphasize domain-invariant spectral components.

Alignment proceeds in two stages.

Global alignment. The first stage minimizes a Similarity Distribution Entropy (SDE) objective over a memory bank containing target features and source prototypes. Cosine similarities define a probability distribution

$$\beta_{i,j} = \frac{\exp(\hat{\mathcal{M}}_j^\top z_i^t / \tau)}{\sum_k \exp(\hat{\mathcal{M}}_k^\top z_i^t / \tau)}, \quad (3.13)$$

whose entropy is minimized according to

$$\mathcal{L}_{\text{GA}} = -\frac{1}{|B|} \sum_{i \in B} \sum_j \beta_{i,j} \log \beta_{i,j}. \quad (3.14)$$

Because all features are L_2 -normalized onto the unit hypersphere, optimization depends primarily on directional consistency rather than feature magnitude, which mitigates gradient instability caused by amplitude discrepancies in frequency features. Shared target samples become attracted toward source prototypes, while target private samples, which exhibit flat similarity distributions where no single prototype dominates, experience repulsive gradients that drive them to form independent clusters.

Local alignment. The second stage refines pseudo labels using entropy regularized optimal transport. The OSAM is defined as:

$$\gamma^* = \arg \max_{\gamma \in \Pi(\mu, \nu)} \sum_{i,j} \gamma_{ij} M_{ij}^{\cosine} + \eta H(\gamma), \quad (3.15)$$

where $M^{\cosine}$ is the cosine similarity matrix between target features and source prototypes, and the marginal constraints $\Pi(\mu, \nu)$ enforce global distributional consistency. This prevents degenerate assignments and produces globally consistent pseudo labels, unlike greedy approaches that assign labels independently for each sample.

Target samples are considered shared only if both conditions hold:

$$\omega_i^t \geq \frac{1}{|B|} \quad \text{and} \quad \omega_{y_i^t}^s \geq \frac{1}{|\mathcal{C}_s|}, \quad (3.16)$$

where $\omega_i^t = \sum_j \bar{\gamma}_{i,j}^*$ measures how strongly target sample i is assigned to any source prototype, and $\omega_j^s = \sum_i \bar{\gamma}_{i,j}^*$ measures how frequently source prototype j is assigned to target samples. Samples violating either condition are rejected as unknown. These adaptive decision boundaries are derived from the dimensions of the transport plan itself rather than fixed hyperparameters.

3.7.2 Partial Domain Adaptation

SPADA (Stacked Autoencoder Based Partial Adversarial Domain Adaptation)

SPADA [32] addresses the partial domain adaptation setting in which

$$\mathcal{C}_t \subset \mathcal{C}_s. \quad (3.17)$$

The architecture combines stacked autoencoders with three classifiers:

1. a weighted domain discriminator D_w ,
2. an adversarial discriminator D_d ,
3. and a source label classifier D_l .

The weighting discriminator estimates

$$D_w^*(f_s(x)) = \frac{p_s(f_s(x))}{p_s(f_s(x)) + p_t(f_s(x))}. \quad (3.18)$$

Source private samples tend to produce discriminator outputs close to one because they are dissimilar to target features. Their importance weight is therefore defined as

$$\tilde{\mu}(f_s(x)) = 1 - D_w^*(f_s(x)). \quad (3.19)$$

Shared class samples receive large weights, while source outlier classes are suppressed during alignment. This weighting operates at the *sample level*: each individual source sample receives its own weight based on how source-like versus target-like its features appear to the discriminator. The weighted features are then used within an adversarial objective implemented using gradient reversal.

PDAAN (Partial Domain Adaptation Adversarial Network)

PDAAN [60] addresses partial domain adaptation through *class-level* weighting combined with adversarial alignment.

The method estimates the importance of each source class by averaging classifier predictions over unlabeled target samples:

$$\omega = \frac{1}{n_t} \sum_{i=1}^{n_t} p_i^t, \quad (3.20)$$

where $p_i^t = C(x_i^t)$ is the softmax output for target sample i . If a source class k is absent from the target domain, few target samples will be predicted as class k , producing a small class-level weight ω^k . These weights are normalized as $\omega \leftarrow \omega / \max(\omega)$ and applied to both a weighted cross-entropy classification loss and a weighted complementary entropy loss. The complementary entropy penalizes prediction uncertainty on classes other than the predicted class, ensuring that fault categories absent from the target domain receive low prediction probability. Additionally, PDAAN expands the target domain with randomly selected source samples during adversarial training to reduce negative transfer caused by label-space asymmetry. The proportion ρ of source samples used for expansion decreases gradually during training as feature alignment improves.

The key distinction from SPADA is the granularity of weighting: SPADA assigns per-sample weights via a dedicated domain discriminator, while PDAAN assigns per-class weights via averaged classifier predictions. SPADA’s sample-level approach is more fine-grained but requires training an additional discriminator network, whereas PDAAN’s class-level approach is simpler and uses only the existing classifier output.

3.7.3 Universal Domain Adaptation

UAN (Universal Adaptation Network)

UAN, introduced by You u. a. [59], was among the first methods proposed specifically for universal domain adaptation.

The architecture consists of:

1. a feature extractor F ,
2. a classifier G ,
3. an adversarial discriminator D ,
4. and a non adversarial discriminator D' .

The discriminator D' estimates domain similarity independently of adversarial optimization. UAN combines this signal with prediction entropy to define transferability scores:

$$w^s(x) = \frac{H(\hat{y})}{\log |\mathcal{C}_s|} - \hat{d}'(x), \quad (3.21)$$

$$w^t(x) = \hat{d}'(x) - \frac{H(\hat{y})}{\log |\mathcal{C}_s|}. \quad (3.22)$$

Samples that appear domain invariant and simultaneously yield confident predictions are considered transferable. These weights modulate the adversarial alignment objective.

At inference time, target samples whose transferability score falls below a threshold $w_0 = 0.5$ are rejected as unknown.

OVANet (One versus All Network)

OVANet [44] approaches unknown rejection through explicit one versus all classification. The model contains:

1. a shared feature extractor F ,
2. a closed set classifier C ,
3. and a one versus all module O .

The OVA module contains one binary classifier per source category. Training uses Hard Negative Classifier Sampling (HNCS), where only the positive classifier and the hardest negative classifier receive updates:

$$\mathcal{L}_{\text{ova}}(x^s, y^s) = -\log p(\hat{y}^{y^s} | x^s) - \min_{j \neq y^s} \log (1 - p(\hat{y}^j | x^s)). \quad (3.23)$$

This encourages tighter class boundaries and improves rejection of unseen categories.

DANCE (Domain Adaptive Neighborhood Clustering via Entropy Optimization)

DANCE [46] combines self supervised target clustering with entropy based separation between shared and private samples.

The classifier weights

$$\{w_1, \dots, w_K\} \quad (3.24)$$

serve as source prototypes. For each target feature f_i , similarities are computed according to

$$p_{i,j} = \frac{\exp(F_j^\top f_i / \tau)}{\sum_{k \neq i} \exp(F_k^\top f_i / \tau)}. \quad (3.25)$$

Entropy minimization encourages target samples either toward source prototypes or toward target specific clusters:

$$-\sum_j p_{i,j} \log p_{i,j}. \quad (3.26)$$

Entropy separation further compares prediction entropy against a fixed threshold

$$\rho = \frac{\log K}{2}. \quad (3.27)$$

The entropy separation loss is defined as

$$\mathcal{L}_{\text{es}}(p) = \begin{cases} -|H(p) - \rho|, & |H(p) - \rho| > m, \\ 0, & \text{otherwise.} \end{cases} \quad (3.28)$$

Notably, ρ is fixed and contains no notion of dataset scale or distributional shift. Any target sample whose softmax entropy exceeds ρ is labelled as unknown, regardless of

whether the high entropy arises from genuine novelty or from low classifier confidence under domain shift.

PPOT (Prototypical Partial Optimal Transport)

PPOT [57] reformulates universal adaptation as a partial optimal transport problem in which only part of the probability mass is transported across domains.

Partial transport constrains transported mass according to

$$\alpha \leq \min(\|\mu\|_1, \|\nu\|_1). \quad (3.29)$$

The mass fraction α is adaptively estimated as the proportion of target samples whose prediction confidence exceeds a threshold τ_1 . When target confidence is uniformly low, as often occurs under cross-dataset transfer, α pins to a hard floor $\alpha_{\min}$, causing the transport plan to become nearly empty.

Source samples are summarized using class prototypes

$$\{c_k\}_{k=1}^{|\mathcal{C}_s|}, \quad (3.30)$$

and transport is solved between source prototypes and target features. The resulting transport plan produces both class level and sample level sharedness weights that guide alignment and unknown rejection.

UniOT (Unified Optimal Transport)

UniOT [9] decomposes universal adaptation into two optimal transport problems:

1. shared class detection across domains,
2. and target representation learning within the target domain.

The first transport problem aligns target features with source prototypes using unbalanced optimal transport with adaptive filling. The unbalanced formulation allows some probability mass to remain untransported, meaning target samples from unknown classes naturally receive low assignment weight without requiring an explicit rejection threshold. The resulting assignment matrix $\mathbf{Q}^{st}$ identifies likely shared samples using assignment confidence and prototype usage statistics.

The second transport problem performs balanced optimal transport between target features and learnable target prototypes. This induces pseudo label distributions that encourage both global target discrimination and local neighborhood consistency.

UniJDOT (Universal Deep Joint Distribution Optimal Transport)

UniJDOT [35] extends DeepJDOT [12] to universal time series adaptation.

The original DeepJDOT alignment cost jointly matches feature and label spaces:

$$C_{ij} = \mu \|G(x_i^s) - G(x_j^t)\|_2^2 + \|y_i^s - h(x_j^t)\|_2^2. \quad (3.31)$$

A key challenge in universal DA is that standard classifiers exhibit overconfidence on unknown samples [29], assigning them high probability for an incorrect known class. To mitigate this, UniJDOT constructs a joint decision space that combines classifier logits with distance-based probabilities computed from a classwise source memory bank:

$$p'_t = \sigma \left(h(x^t) \odot \sigma(-d_t) \right), \quad (3.32)$$

where $d_t = (\min_{x^s \in \mathcal{X}_s^1} d(x^t, x^s), \dots, \min_{x^s \in \mathcal{X}_s^K} d(x^t, x^s))$ is a vector of distances from the target sample to its nearest source neighbor in each class. The elementwise product ensures that a high classifier score is suppressed if the sample lies far from the corresponding source class in feature space, thereby increasing the indecision margin for unknown samples.

The maximum values of $\{p'_t\}$ within each batch are assembled into a histogram, and a binary auto-thresholding algorithm (Yen’s method [58]) determines the separation between common and unknown samples without requiring a fixed or fine-tuned hyperparameter. This batch-adaptive thresholding is a central design choice: the threshold adjusts to the current batch distribution rather than relying on global constants calibrated on a different dataset.

Unknown target samples are transported toward feature space anchors $\{a_l\}_{l=1}^L$ (computed as K -means centroids of target features, updated via moving average) and a maximally uncertain prediction vector

$$r = \frac{1}{|\mathcal{C}_s|} \mathbf{1}. \quad (3.33)$$

The resulting unknown transport cost becomes

$$C_{kj}^{\mathcal{U}} = \mu \|a_k - G(x_j^t)\|_2^2 + \|r - h(x_j^t)\|_2^2. \quad (3.34)$$

Common target samples are aligned with source samples using the standard DeepJDOT cost (Equation above), while unknown target samples are aligned with anchors using $C^{\mathcal{U}}$, within a unified cost matrix solved via unbalanced optimal transport.

RAINCOAT (Frequency Augmented Align then Correct for Domain Adaptation for Time Series)

RAINCOAT [20] is specifically designed for universal domain adaptation on time series data. The method models both temporal and spectral domain shift.

The feature extractor contains:

1. a CNN branch producing temporal features $\mathbf{e}_T$,
2. and a frequency branch producing amplitude and phase features.

The final representation is

$$\mathbf{z} = [\mathbf{e}_T; \mathbf{a}; \mathbf{p}]. \quad (3.35)$$

During the *align* stage, source and target features are matched using Sinkhorn divergence. A source reconstruction loss $\mathcal{L}_R = |x^s - U_{\text{TF}}(G_{\text{TF}}(x^s))|$ is jointly optimized to promote learning of semantic features. During the *correct* stage, the encoder and decoder are retrained on target data only through a reconstruction objective $\mathcal{L}_R = |x^t - U_{\text{TF}}(G_{\text{TF}}(x^t))|$, which allows target private samples to drift away from their forced alignment to source clusters.

The method detects unknown samples by measuring the absolute difference in cosine distance to the assigned prototype before and after the correction step:

$$d_i^{ac} = |d(\mathbf{z}_{a,i}^t, \mathbf{w}_c) - d(\mathbf{z}_{c,i}^t, \mathbf{w}_c)|, \quad (3.36)$$

where $\mathbf{z}_{a,i}^t$ and $\mathbf{z}_{c,i}^t$ are the target features before and after correction, and $\mathbf{w}_c$ is the assigned class prototype. Common-class samples, which were correctly aligned during the first stage, move less during correction than unknown-class samples, whose forced alignment gets undone. The resulting distribution of d^{ac} is tested for bimodality using the Hartigan dip test. If bimodal structure is detected ($p < 0.05$), two-means clustering separates known from unknown samples, with the larger centroid serving as the rejection threshold.

4 Results

This chapter presents the experimental results for the closed-set, OSDA, PDA, and UniDA scenarios. Before looking at the performance of all the algorithms across our scenarios, it's good to keep in mind the significant difference in the training times. We find in the diagram 4.1 that UniOT and UniJDOT can take up to 30 minutes when the datasets are large.

Mean training duration per algorithm — one panel per pair (averaged across scenarios, n_{unknown} , seeds)

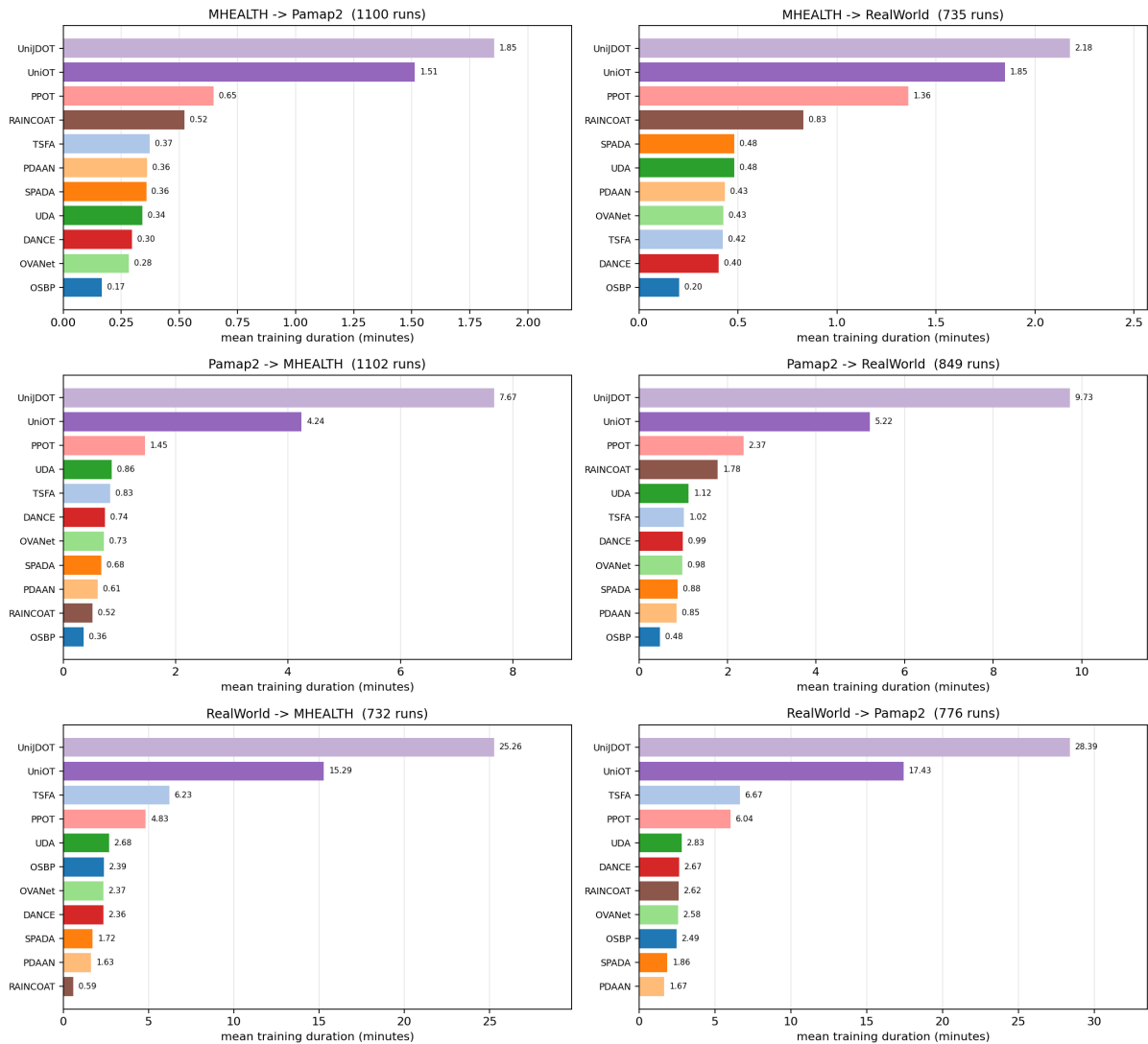


Abbildung 4.1: Mean duration of training for all algorithms for each pair of datasets

4.1 Experiment 1: Closed Set Scenario Across All Methods

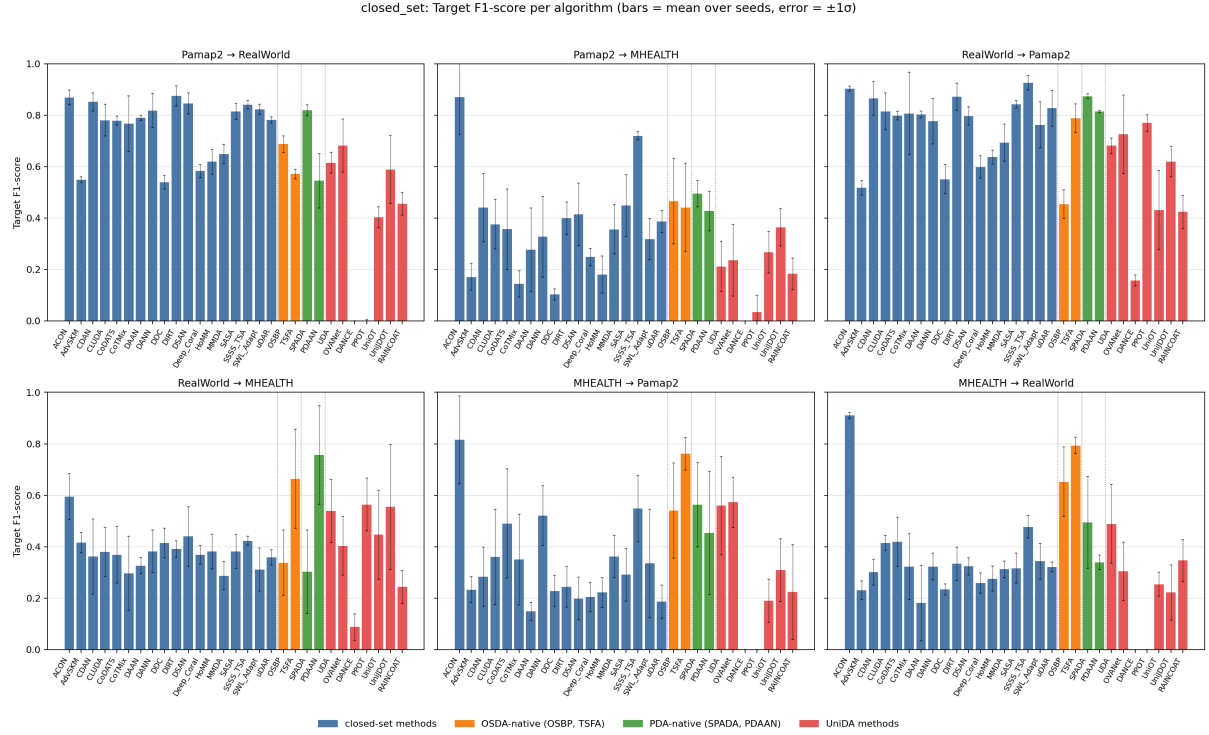


Abbildung 4.2: Closed-set target Macro F1-score comparison across methods.

4.2 Experiment 2: OSDA Scenario

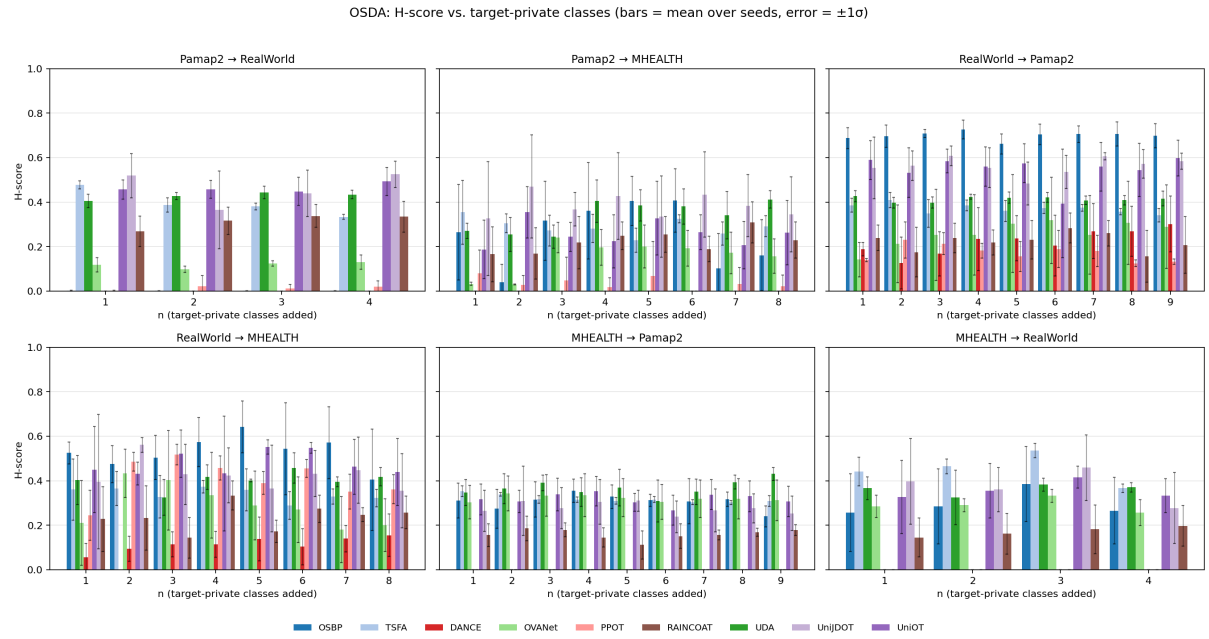


Abbildung 4.3: OSDA H-score comparison across methods.

4.3 Experiment 3: PDA Scenario

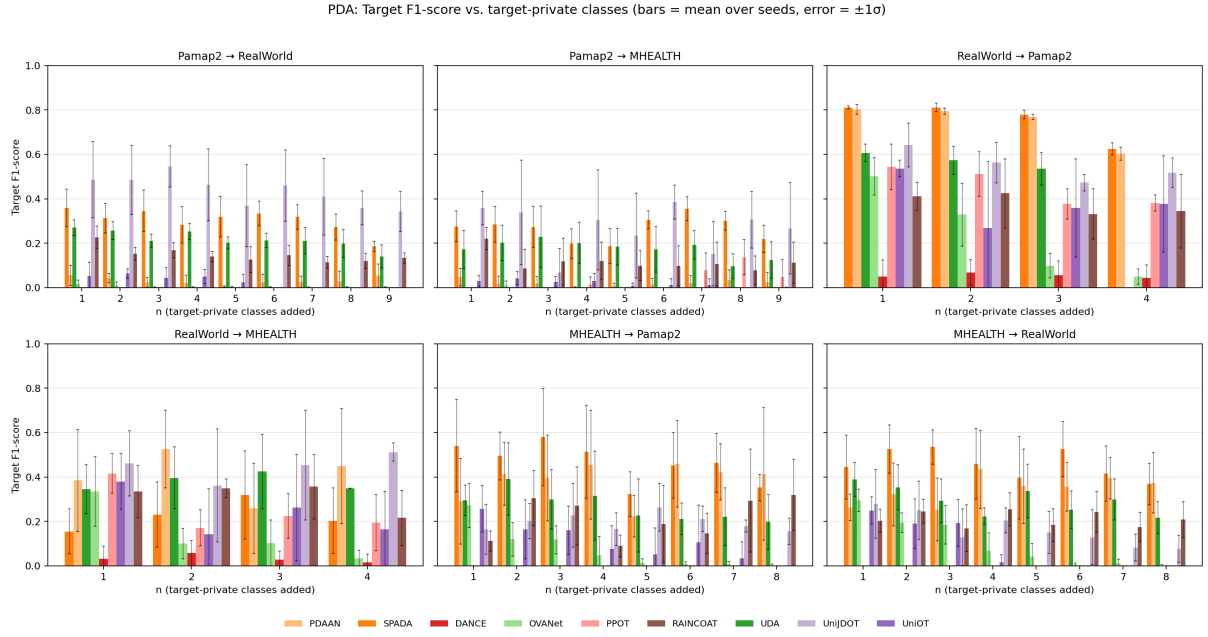


Abbildung 4.4: PDA target F1 comparison across methods.

4.4 Experiment 4: UniDA Scenario

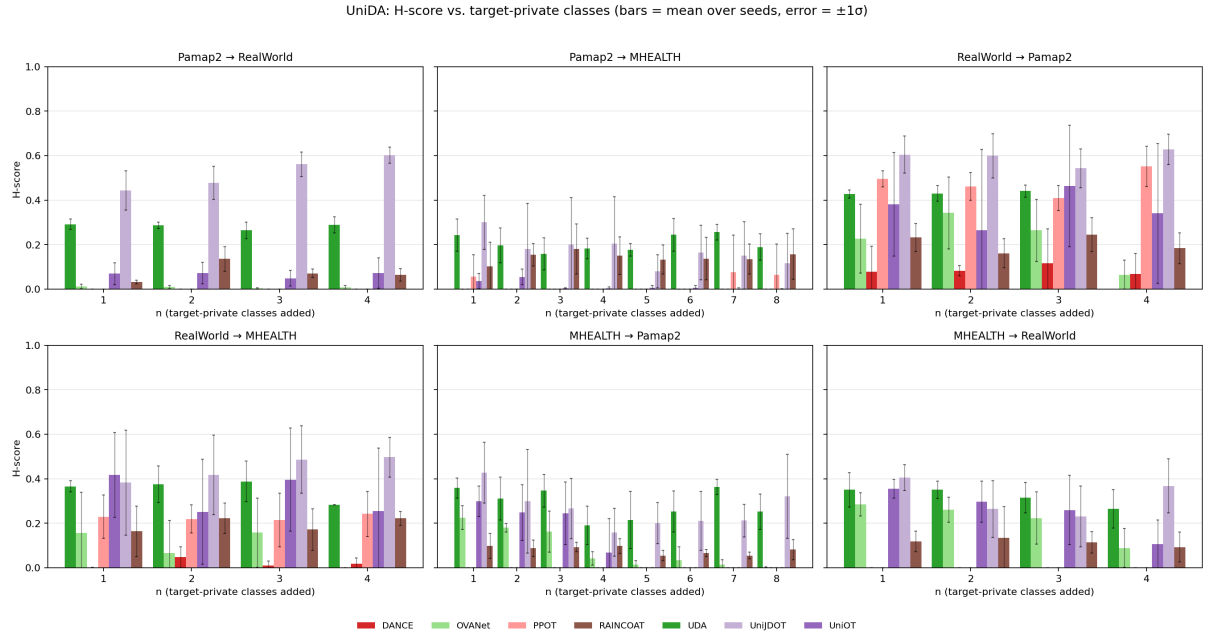


Abbildung 4.5: UniDA H-score comparison across methods.

5 Discussion

5.1 Phase 1 : Closed-Set Scenario Across All Methods

5.1.1 Headline Results

Setting aside ACON, which dominates with mean target $F1 = 0.83$ and tops five of six dataset pairs, the remainder of the closed-set-designed family is remarkably tightly clustered. The next eleven closed-set methods lie between $F1 = 0.42$ (DAAN) and 0.66 (SSSS-TSA), with most concentrated in the 0.48 – 0.53 range (DANN 0.52 , CLUDA 0.52 , DIRT-T 0.52 , CDAN 0.52 , SASA 0.52 , DSAN 0.50). The family-wide mean is 0.49 ± 0.25 . The UniDA family averages 0.34 ± 0.25 , approximately 15 points lower. However, the top UniDA method, UDA at 0.52 , matches the closed-set median. Consequently, the statement that “UniDA loses to closed-set” is driven primarily by a subset of collapsing UniDA methods rather than by a uniform deficit across the family.

Source-side accuracy remains nearly identical across both families (0.94 vs 0.92), implying that the performance gap is paid entirely on the transfer side.

5.1.2 Dataset Size Effects

Dataset size influences transfer performance before algorithmic differences become dominant. RealWorld is the largest dataset of the three, with approximately 41,900 labelled windows across eight activities. Pamap2 contains approximately 19,100 windows across thirteen non-empty classes, while MHEALTH contains approximately 4,600 windows across twelve classes. RealWorld is therefore approximately nine times larger than MHEALTH and roughly twice the size of Pamap2.

The closed-set performance gap is not uniform across dataset pairs. When RealWorld appears as either source or target, closed-set methods reach $F1 \approx 0.75$ – 0.77 . When MHEALTH appears on either side, nearly all methods drop below $F1 < 0.40$, and the family gap narrows to only a few points. Larger target manifolds therefore appear to benefit discriminative closed-set classifiers while simultaneously exposing unnecessary unknown-rejection machinery in UniDA methods.

5.1.3 Why DANCE and PPOT Collapse

DANCE

DANCE averages $F1 = 0.04$ with peak performance only reaching 0.18 , constituting near-total collapse. The mechanism is built around a fixed entropy threshold $\rho = \frac{1}{2} \log K$. Any target sample whose softmax entropy exceeds ρ is labelled as unknown. Under closed-set transfer, every target label belongs to the shared label space, meaning every rejection is necessarily a false rejection.

On smaller datasets, especially MHEALTH-source or MHEALTH-target pairs, feature extractor confidence is intrinsically lower. Average entropy therefore remains above ρ ,

causing the entire target stream to be labelled unknown and yielding F1 scores near zero. Only the largest-source pairs (RealWorld $\rightarrow$ Pamap2 and RealWorld $\rightarrow$ MHEALTH) produce sufficient classifier confidence to permit some correct predictions, scoring 0.16 and 0.09 respectively.

The fixed threshold ρ contains no notion of dataset scale or distributional shift. Furthermore, the selected hyperparameters $\eta = 0.05$ and margin = 0.95 were tuned on RealWorld male $\rightarrow$ female transfer and further amplify the issue by aggressively separating entropy distributions around ρ .

PPOT

PPOT exhibits the strongest variance in the benchmark, with mean F1 = 0.23 and standard deviation = 0.32, alongside a clear bimodal pattern. It achieves 0.77 on RealWorld $\rightarrow$ Pamap2 and 0.56 on RealWorld $\rightarrow$ MHEALTH while collapsing to approximately zero on all remaining pairs.

The mechanism depends on adaptive estimation of the partial-OT mass α , defined as the fraction of target samples whose confidence exceeds $\tau_1 = 0.55$. On RealWorld-source pairs this fraction remains well above the hard floor $\alpha_{\min} = 0.05$, allowing the Sinkhorn transport plan to allocate meaningful mass. On Pamap2-source and MHEALTH-source pairs, however, target confidence rarely exceeds τ_1 , causing α to pin to its minimum value. The resulting transport plan becomes nearly empty, class weights collapse toward zero, and the weighted source cross-entropy degenerates into noise.

Again, $\tau_1 = 0.55$ was selected on RealWorld male $\rightarrow$ female transfer, where confidence distributions are substantially cleaner than under cross-dataset HAR transfer.

5.1.4 UniDA Methods That Survive

The remaining UniDA methods do not collapse: UDA (0.52), OVANet (0.49), UniJDOT (0.44), UniOT (0.33), and RAINCOAT (0.31). UDA in particular performs within a few points of the closed-set median across all six dataset pairs. Mechanistically, UDA’s GRL-based alignment behaves similarly to a conventional closed-set DA method such as DANN rather than a strict unknown-rejection framework.

The key distinction is therefore not “UniDA versus closed-set” but rather learned rejection versus fixed-threshold rejection. Methods whose rejection mechanisms are learned from data (UDA, OVANet, UniJDOT) remain robust, while methods relying on hard-coded priors (DANCE, PPOT) collapse under distribution shift.

5.1.5 Native OSDA and PDA Methods on Closed-Set

The four scenario-native methods are not part of the closed-set baseline family, but they share its assumption when the label space happens to coincide on both sides. Under that regime they cluster between OSBP = 0.52 and TSFA = 0.67, with PDAAN = 0.56 and SPADA = 0.59 in between. The OSDA-native methods averages 0.60 ± 0.18 and the PDA-native methods 0.57 ± 0.22 , both above the closed-set family mean of 0.49 and substantially above the UniDA mean of 0.34.

Mechanistically, none of these methods rely on a fixed entropy threshold or a confidence-pinned mass estimator. OSBP’s $t = 0.5$ boundary degenerates into a vacuous binary decision when no class is private, leaving its GRL-based source-target alignment to dominate. Essentially a DANN variant with an unused head. SPADA’s source-importance

weights converge towards uniform under matched label spaces, since no class is genuinely down-weighted; PDAAN’s class-aware adversary likewise behaves like a per-class CDAN. TSFA’s temporal-shift augmentation operates entirely on the source side and provides closed-set generalisation gain regardless of label-space, explaining why it produces the strongest non-ACON F1 on RealWorld $\rightarrow$ MHEALTH (0.66), RealWorld $\rightarrow$ Pamap2 (0.79) and the MHEALTH-source pairs (0.76 and 0.79).

The per-pair pattern mirrors the closed-set family. SPADA and PDAAN earn their averages from RealWorld-source pairs (SPADA = 0.87 on RealWorld $\rightarrow$ Pamap2, PDAAN = 0.81) and lose them when MHEALTH appears as the target (SPADA = 0.30 on RealWorld $\rightarrow$ MHEALTH, PDAAN = 0.34 on MHEALTH $\rightarrow$ RealWorld). The main observation is that label-space-mismatch methods, when run in the closed-set scenario, behave as their alignment backbone alone; the rejection machinery becomes inert rather than harmful.

5.1.6 Hyperparameter Notice

All UniDA methods in this phase use hyperparameters selected from a RealWorld male $\rightarrow$ female within-dataset sweep. PPOT’s $\tau_1 = 0.55$, DANCE’s $\eta = 0.05$, and various OT regularisation constants were therefore calibrated against a target distribution approximately nine times larger than MHEALTH.

Methods such as UniJDOT, UDA, and OVANet survive this transposition because their objectives do not depend on fixed thresholds over raw target confidence. PPOT and DANCE do.

5.1.7 Take-Away

The closed-set phase yields three central findings. First, in the absence of private classes, closed-set-designed methods retain an average advantage of approximately 15 F1 points over the UniDA family, with ACON emerging as the dominant single method. Second, the OSDA-native and PDA-native methods, whose rejection machinery is inert under matched label spaces, average above the closed-set family itself (0.60 and 0.57 vs 0.49), confirming that the closed-set vs UniDA gap is not about extra unknown-handling heads being added but about *how rejection is parameterised*. Third, this average masks a sharp split within the UniDA family: methods with learned rejection mechanisms (UDA, OVANet, UniJDOT) match the closed-set median, while methods with fixed-threshold rejection (DANCE, PPOT) collapse catastrophically under confidence-distribution shift.

5.2 Phase 2: Open-set Scenario

5.2.1 Headline Results

The OSDA scenario changes the ranking structure observed in closed-set transfer. Unlike Phase 1, there is no dominant method equivalent to ACON. Instead, the top of the leaderboard compresses into a narrow cluster: UniJDOT (0.41), UniOT (0.40), OSBP (0.39), UDA (0.38), and TSFA (0.35). All five methods lie within seven points of each other and frequently overlap within seed variance.

The native OSDA family averages $H = 0.37$, while the UniDA family averages $H = 0.26$. However, this gap is entirely explained by DANCE (0.07) and PPOT (0.12). Ex-

cluding those two methods, the remaining UniDA methods average $H = 0.33$, which is effectively indistinguishable from the native OSDA methods.

Consequently, once unknown target classes actually exist, rejection machinery ceases to be a liability. Methods with learnable rejection mechanisms (UniJDOT, UniOT, UDA, OVANet) catch up to dedicated OSDA methods.

5.2.2 OS* / UNK Decomposition

The harmonic mean $H = HM(OS^*, UNK)$ conceals strong behavioural differences between methods. OSBP is the only genuinely balanced method, reaching $OS^* = 0.55$ and $UNK = 0.43$. TSFA and OVANet achieve strong known-class classification but under-reject unknowns. UniJDOT and UniOT exhibit the opposite behaviour, prioritising rejection over known-class accuracy. DANCE effectively predicts almost everything as unknown, reaching $UNK = 0.96$ while collapsing to $OS^* = 0.04$.

The contrast with the closed-set scenario becomes immediately interpretable. Under closed-set transfer, DANCE collapsed because every rejection was false. Under OSDA, those same rejections partially become correct, increasing H-score despite no actual improvement in known-class classification. Conversely, TSFA and OVANet now appear weaker because their rejection mechanisms are overly conservative.

5.2.3 Behaviour Along the Increasing Unknown-Class Strategy

Linear trends across increasing numbers of target-private classes reveal three behavioural classes. OSBP exhibits the strongest positive slope, increasing by approximately +0.017 H-score per added unknown class, climbing from $H = 0.34$ at $n = 1$ to $H = 0.51$ at $n = 5$ before plateauing. This behaviour emerges because OSBP’s adversarial target prior assumes a binary known/unknown distribution centred near $(0.5, 0.5)$. As n increases, the actual target composition increasingly approximates this prior.

UniJDOT, UDA, UniOT, OVANet, and RAINCOAT all exhibit nearly flat slopes with respect to n . Their mechanisms rely on continuous prototype distances, GRL-based alignment, or adaptive transport costs rather than on explicit dependence upon the number of unknown classes. TSFA instead degrades steadily, falling from $H = 0.40$ at $n = 1$ to $H = 0.32$ at $n = 9$ because its fixed rejection threshold becomes increasingly under-calibrated as the unknown fraction grows.

5.2.4 Variability and Robustness

The most stable methods by within-cell seed variance are DANCE, PPOT, TSFA, and UDA. However, DANCE and PPOT are merely stable near zero. Among genuinely competitive methods, UDA and TSFA are the most consistent. UniJDOT trades stability for ceiling performance, achieving the highest single H-score in the benchmark ($H = 0.72$ on RealWorld $\rightarrow$ Pamap2 at $n = 4$).

5.2.5 Take-Away

In closed-set transfer, rejection machinery imposed unnecessary cost. In OSDA, rejection becomes essential. Methods with adaptive or learned rejection (UniJDOT, UniOT, UDA,

OSBP) become competitive with native OSDA methods, while methods relying on fixed confidence thresholds (DANCE, PPOT) remain non-functional.

The central distinction is therefore adaptive rejection versus fixed-prior rejection.

5.3 Phase 3 : Partial-set Scenario

5.3.1 Headline Results

PDA reverses the trend observed in OSDA and strongly favours native PDA methods. SPADA leads the benchmark at mean target F1 = 0.39 and is the only method remaining above 0.30 across the full experiment. UniJDOT is the strongest UniDA method at 0.31, followed by UDA at 0.27 and PDAAN at 0.26.

The remaining UniDA methods collapse: UniOT (0.11), PPOT (0.08), OVANet (0.07), and DANCE (0.01). The family averages become native PDA = 0.33 and UniDA = 0.15, producing an 18-point deficit larger than both the closed-set and OSDA gaps.

5.3.2 Why PDA Reverses the OSDA Trend

OSDA and PDA require fundamentally different reasoning. OSDA asks which target samples have no source class, whereas PDA asks which source classes have no target support. Most UniDA methods were explicitly designed for the former. Consequently, adding source-private classes introduces distractor classes that pull feature alignment toward regions absent from the target distribution, producing strong negative transfer.

Slope analysis confirms this interpretation. All methods degrade as n increases, but degradation is steepest for methods lacking explicit source-class weighting: UniOT (-0.031), UDA (-0.027), and OVANet (-0.025).

5.3.3 SPADA vs PDAAN

SPADA employs two discriminators: a weighting discriminator assigning per-sample weights $\tilde{\omega} = 1 - \sigma(D_w(f_s))$, and a separate GRL-based alignment discriminator. This produces fine-grained suppression of source-private samples. Samples appearing distinctly source-private receive near-zero weight and contribute negligibly to either alignment or classification loss. The resulting performance curve remains relatively flat across most of the unknown increase and only degrades sharply at extreme n values.

PDAAN instead operates at the class level through an EMA-updated class-weight buffer $\omega \in \mathbb{R}^K$. This mechanism is conceptually elegant but reacts too slowly at large n . As the number of source-private classes increases, the EMA cannot converge rapidly enough before training terminates. Entropy-based regularisation then amplifies the imbalance, causing target predictions to collapse onto a small subset of classes. The resulting behaviour is characteristic: 0.31 at $n = 1$ but only 0.039 at $n = 9$.

5.3.4 UniDA Methods Under PDA

UDA lacks any PDA-specific weighting mechanism. It attempts to align all source classes equally, including source-private distractors, causing target F1 to degrade approximately linearly from 0.35 to 0.13 across the curriculum.

UniJDOT becomes the strongest UniDA method because its distance-based transport implicitly suppresses source-private prototypes. Transport toward unmatched source prototypes incurs large cost and therefore receives negligible weight. Mechanistically, this resembles SPADA’s weighting strategy but operates at the prototype level rather than the sample level.

UniOT collapses catastrophically from 0.25 at $n = 1$ to 0.00 by $n = 8$. The inferred target marginal becomes smeared across source-private classes, invalidating the assumptions underlying prior-free OT estimation.

OVANet’s independent one-vs-all heads become negatively biased by source-private classes. Heads corresponding to absent target classes learn to output “not class k ” for the entire target distribution, reducing aggregate confidence and inducing rejection collapse.

RAINCOAT is the most resilient non-PDA-specific method. Its frequency-domain drift detection does not directly suppress source-private classes but also avoids strongly amplifying them.

5.3.5 Take-Away

PDA reveals the limits of “universal” adaptation. OSDA-oriented rejection mechanisms do not naturally transfer to PDA, and methods survive only if they possess implicit source-awareness or weighting. Prototype-level and sample-level weighting mechanisms therefore dominate threshold-based rejection. The strongest surviving methods under PDA are SPADA, PDAAN, and UniJDOT.

5.4 Phase 4: Universal Scenario

5.4.1 Headline Results

UniDA is the most difficult phase in the benchmark. Mean H-scores compress downward across all methods: UniJDOT (0.35), UDA (0.29), UniOT (0.15), RAINCOAT (0.13), PPOT (0.09), OVANet (0.08), and DANCE (0.01). No method exceeds a global mean H-score of 0.50. UniJDOT is the only method achieving a per-pair mean above that threshold, reaching $H = 0.59$ on RealWorld $\rightarrow$ Pamap2.

5.4.2 Behaviour Along the Unknown Increase

Every method exhibits a negative slope with increasing n . UniJDOT and UniOT exhibit the steepest declines (-0.040), but through different mechanisms. UniJDOT degrades gradually from 0.43 to 0.22 while remaining the best method at every n . UniOT instead collapses entirely from 0.23 to 0.00 by $n = 5$.

UDA and RAINCOAT are comparatively resilient because neither relies heavily upon explicit rejection thresholds.

5.4.3 OS* / UNK Decomposition

UniDA amplifies the rejection imbalance already visible in OSDA. UNK rates become extremely high: DANCE (0.94), PPOT (0.80), UniOT (0.71), and UniJDOT (0.66), while OS* remains uniformly low: UniJDOT (0.29), UDA (0.26), and UniOT (0.11).

The structural reason is straightforward: source-private classes degrade posterior confidence on shared target classes, and every explicit rejection mechanism interprets uncertainty as evidence for “unknown”. UDA, lacking explicit rejection entirely, exhibits the most balanced OS*/UNK trade-off.

5.4.4 Why UniJDOT and UDA Survive

UniJDOT is the only method naturally handling both source-private and target-private mismatch through the same primitive: joint distance-based transport. Source-private prototypes incur large transport cost because no nearby target features exist, while target-private samples similarly fail to align with source prototypes. Adaptive thresholding then operates on current-batch statistics rather than fixed global priors. This makes UniJDOT relatively insensitive to hyperparameter transposition.

UDA survives precisely because it never attempts explicit rejection. Its threshold-free GRL objective remains stable under both PDA and OSDA mismatch simultaneously. This sacrifices unknown detection performance but preserves shared-class classification.

5.4.5 Why the Other Methods Fail

UniOT’s prior-free OT assumption breaks once both source-private and target-private classes coexist. The inferred marginal becomes increasingly diffuse, transport plans approach uniformity, and decoding defaults toward the unknown class.

OVANet’s OVA heads become negatively biased by source-private classes, reducing aggregate confidence and driving rejection rates upward.

DANCE and PPOT preserve the same failure modes identified in earlier phases: DANCE fails due to fixed entropy thresholding, while PPOT fails due to confidence-threshold collapse of the partial OT mass.

RAINCOAT’s drift detector is in principle direction-agnostic and therefore theoretically UniDA-compatible. However, the drift test remains too conservative to robustly separate shared and private structure under cross-dataset HAR transfer.

5.4.6 Cross-Phase Comparison

Method	Closed-Set	OSDA	PDA	UniDA
UDA	0.52	0.38	0.27	0.29
OVANet	0.49	0.25	0.07	0.08
DANCE	0.04	0.07	0.01	0.01
PPOT	0.23	0.12	0.08	0.09
UniOT	0.33	0.40	0.11	0.15
UniJDOT	0.44	0.41	0.31	0.35
RAINCOAT	0.31	0.22	0.20	0.13

UniJDOT is the only method whose performance degrades smoothly without catastrophic collapse across all four settings.

6 Conclusion

This thesis introduced a unified cross-paradigm benchmark of domain adaptation methods on time-series Human Activity Recognition, covering closed-set, OSDA, PDA and UniDA under identical training and evaluation protocols. We re-implemented OSBP, TS-FA, SPADA and PDAAN for time-series and combined them with seven UniDA methods and eighteen closed-set baselines, then probed robustness with a hardest-first strategy over private-class counts.

Three findings define the picture. *First*, under matched label spaces the closed-set vs UniDA gap is not caused by extra unknown-handling heads: OSDA-native and PDA-native methods, whose rejection machinery is inert when no private classes exist, outperform the closed-set family on average. *Second*, once private classes appear, the decisive distinction is adaptive versus fixed-threshold rejection. Methods relying on global priors (DANCE’s ρ , PPOT’s τ_1 , fixed OT priors, OVA thresholds) were all tuned on a RealWorld male $\rightarrow$ female sweep and collapse under cross-dataset confidence shift, while UniJDOT, UDA, OSBP and OVAANet remain competitive. *Third*, the UniDA phase establishes an effective ceiling of $H \approx 0.35$, with UniJDOT, which replaces fixed thresholds with adaptive batch-wise distance gating, and UDA, which avoids thresholding entirely, the only methods degrading gracefully across all four phases.

Together these results align with UniDABench: successful UniDA for time-series transfer must be either threshold-free or adaptive to the current batch distribution, never reliant on a global prior carried over from a simpler transfer scenario.

Two directions stand out for future work. The first is novel class discovery: rather than only flagging target-private samples as “unknown”, future work should cluster them into their underlying latent activities and assign meaningful labels. A harder problem than open-set rejection but the practically useful continuation. The second is strengthening the partial-set mechanisms inside UniDA methods: the native PDA methods (SPADA, PDAAN) retain a clear lead in the PDA phase precisely because they explicitly down-weight or re-weight source-private classes, whereas most UniDA methods inherit only an OSDA-style rejection head. Borrowing SPADA-style per-sample weighting or PDAAN-style class-level re-weighting into a UniDA objective is therefore a concrete path toward closing the remaining gap on the partial side without losing open-set performance.

7 ACKNOWLEDGMENT

I have used Claude Code to facilitate the development of the experiments framework. I have also used Google Gemini for rewording some of my paragraphs, and the Deep Research feature to search for the most up-to-date references. I have used these tools with the full knowledge of the mistakes they produce regularly, and I verified and tested all the information and functionality produced by them.

Literatur

- [1] Davide Anguita u. a. „A Public Domain Dataset for Human Activity Recognition Using Smartphones“. In: *European Symposium on Artificial Neural Networks*. Bd. 3. 2013, S. 3–4.
- [2] Oresti Banos u. a. „mHealthDroid: A Novel Framework for Agile Development of Mobile Health Applications“. In: *Lecture Notes in Computer Science* 8868 (2014), S. 91–98.
- [3] Shai Ben-David u. a. „A Theory of Learning from Different Domains“. In: *Machine Learning* 79.1–2 (2010), S. 151–175.
- [4] Silvia Bucci, Mohammad Reza Loghmani und Tatiana Tommasi. „On the Effectiveness of Image Rotation for Open Set Domain Adaptation“. In: *ECCV*. 2020, S. 422–438.
- [5] Andreas Bulling, Ulf Blanke und Bernt Schiele. „A Tutorial on Human Activity Recognition Using Body-Worn Inertial Sensors“. In: *ACM Computing Surveys* 46.3 (2014), S. 1–33.
- [6] Ruichu Cai u. a. „Time Series Domain Adaptation via Sparse Associative Structure Alignment“. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Bd. 35. 2021, S. 6859–6867.
- [7] Zhangjie Cao u. a. „Partial Adversarial Domain Adaptation“. In: *ECCV*. 2018, S. 135–150.
- [8] Zhangjie Cao u. a. „Learning to Transfer Examples for Partial Domain Adaptation“. In: *IEEE/CVF CVPR*. 2019, S. 2985–2994.
- [9] Wanxing Chang u. a. „Unified Optimal Transport Framework for Universal Domain Adaptation“. In: *NeurIPS*. Bd. 35. 2022, S. 29512–29524.
- [10] Nicolas Courty u. a. „Joint Distribution Optimal Transportation for Domain Adaptation“. In: *Advances in Neural Information Processing Systems*. Bd. 30. 2017.
- [11] Marco Cuturi. „Sinkhorn Distances: Lightspeed Computation of Optimal Transport“. In: *Advances in Neural Information Processing Systems*. Bd. 26. 2013.
- [12] Bharath Bhushan Damodaran u. a. „DeepJDOT: Deep Joint Distribution Optimal Transport for Unsupervised Domain Adaptation“. In: *European Conference on Computer Vision*. 2018, S. 447–463.
- [13] Emadeldeen Eldele u. a. „Contrastive Domain Adaptation for Time-Series via Temporal Mixup“. In: *IEEE Trans. Artificial Intelligence* 5.3 (2023), S. 1185–1194.
- [14] Zhen Fang u. a. „Open Set Domain Adaptation: Theoretical Bound and Algorithm“. In: *IEEE Transactions on Neural Networks and Learning Systems* 32.10 (2021), S. 4309–4322.
- [15] Jean Feydy u. a. „Interpolating Between Optimal Transport and MMD Using Sinkhorn Divergences“. In: *AISTATS*. 2019.

- [16] Muhammad Tanzil Furqon, Mahardhika Pratama und Wisnu Jatmiko. „Time Series Domain Adaptation: A Review“. In: *arXiv preprint* (2025).
- [17] Yaroslav Ganin u. a. „Domain-Adversarial Training of Neural Networks“. In: *Journal of Machine Learning Research* 17.59 (2016), S. 1–35.
- [18] Reza Ghaffari u. a. „Toward Domain Adaptation with Open-Set Target Data: Review of Theory and Computer Vision Applications“. In: *Information Fusion* 100 (2023), S. 101912.
- [19] Muhammad Ghifary u. a. „Deep Reconstruction-Classification Networks for Unsupervised Domain Adaptation“. In: *ECCV*. 2016, S. 597–613.
- [20] Huan He u. a. „Domain Adaptation for Time Series Under Feature and Label Shifts“. In: *ICML*. 2023, S. 12746–12774.
- [21] Judy Hoffman u. a. „Simultaneous Deep Transfer Across Domains and Tasks“. In: *IEEE International Conference on Computer Vision*. 2015, S. 4068–4076.
- [22] Hassan Ismail Fawaz u. a. „Deep Unsupervised Domain Adaptation for Time Series Classification: A Benchmark“. In: *Data Mining and Knowledge Discovery* 39.4 (2025).
- [23] Joonhyuk Jang u. a. „Unknown-Aware Domain Adversarial Learning for Open-Set Domain Adaptation“. In: *NeurIPS*. Bd. 35. 2022, S. 16755–16767.
- [24] I-Hong Jhuo u. a. „Robust Visual Domain Adaptation with Low-Rank Reconstruction“. In: *IEEE CVPR*. 2012, S. 2168–2175.
- [25] Charmi Jobanputra, Jatna Bavishi und Nishant Doshi. „Human Activity Recognition: A Survey“. In: *Procedia Computer Science*. Bd. 155. 2019, S. 698–703.
- [26] Guoliang Kang u. a. „Contrastive Adaptation Network for Unsupervised Domain Adaptation“. In: *IEEE/CVF CVPR*. 2019, S. 4893–4902.
- [27] Al Kfari und Stefan Lüdtker. „Domain Adaptation in Human Activity Recognition through Self-Training“. In: *Proceedings of the ACM International Joint Conference on Pervasive and Ubiquitous Computing*. 2024, S. 462–468. DOI: 10.1145/3675094.3678465.
- [28] Jennifer R. Kwapisz, Gary M. Weiss und Samuel A. Moore. „Activity Recognition Using Cell Phone Accelerometers“. In: *ACM SigKDD Explorations Newsletter* 12.2 (2011), S. 74–82.
- [29] Balaji Lakshminarayanan, Alexander Pritzel und Charles Blundell. „Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles“. In: *Advances in Neural Information Processing Systems*. Bd. 30. 2017.
- [30] Zongyi Li u. a. „Fourier Neural Operator for Parametric Partial Differential Equations“. In: *ICLR*. 2021.
- [31] Qiao Liu und Hui Xue. „Adversarial Spectral Kernel Matching for Unsupervised Time Series Domain Adaptation“. In: *IJCAI*. 2021, S. 2744–2750.
- [32] Zhao-Hua Liu u. a. „A Stacked Auto-Encoder Based Partial Adversarial Domain Adaptation Model for Intelligent Fault Diagnosis of Rotating Machines“. In: *IEEE Transactions on Industrial Informatics* 17.10 (2021), S. 6798–6809.
- [33] Mingsheng Long u. a. „Learning Transferable Features with Deep Adaptation Networks“. In: *International Conference on Machine Learning*. 2015, S. 97–105.

- [34] Mingsheng Long u. a. „Conditional Adversarial Domain Adaptation“. In: *Advances in Neural Information Processing Systems*. Bd. 31. 2018.
- [35] Romain Mussard u. a. „Deep Joint Distribution Optimal Transport for Universal Domain Adaptation on Time Series“. In: *arXiv preprint arXiv:2503.11217* (2025).
- [36] Romain Mussard u. a. „Universal Domain Adaptation Benchmark for Time Series Data Representation“. In: *EUSIPCO*. 2025.
- [37] Yilmazcan Ozyurt, Stefan Feuerriegel und Ce Zhang. „Contrastive Learning for Unsupervised Domain Adaptation of Time Series“. In: *The Eleventh International Conference on Learning Representations*. 2023.
- [38] Sanqing Qu u. a. „GLC++: Source-Free Universal Domain Adaptation through Global-Local Clustering and Contrastive Affinity Learning“. In: *IEEE TPAMI* 47.11 (2024), S. 10646–10663.
- [39] Mohamed Ragab u. a. „AdaTime: A Benchmarking Suite for Domain Adaptation on Time Series Data“. In: *ACM Transactions on Knowledge Discovery from Data* 17.8 (2023), S. 1–18.
- [40] Ievgen Redko u. a. „Optimal Transport for Multi-Source Domain Adaptation under Target Shift“. In: *AISTATS*. 2019, S. 849–858.
- [41] Attila Reiss und Didier Stricker. „Introducing a New Benchmarked Dataset for Activity Monitoring“. In: *International Symposium on Wearable Computers*. 2012, S. 108–109.
- [42] Daniel Roggen, Alberto Calatroni, Mirco Rossi u. a. „Collecting Complex Activity Datasets in Highly Rich Networked Sensor Environments“. In: *International Conference on Networked Sensing Systems*. 2010, S. 233–240.
- [43] Artem Rozantsev, Mathieu Salzmann und Pascal Fua. „Beyond Sharing Weights for Deep Domain Adaptation“. In: *IEEE TPAMI* 41 (2019), S. 801–814.
- [44] Kuniaki Saito und Kate Saenko. „OVANet: One-vs-All Network for Universal Domain Adaptation“. In: *IEEE/CVF ICCV*. 2021, S. 8980–8989.
- [45] Kuniaki Saito u. a. „Open Set Domain Adaptation by Backpropagation“. In: *ECCV*. 2018.
- [46] Kuniaki Saito u. a. „Universal Domain Adaptation through Self Supervision“. In: *NeurIPS*. Bd. 33. 2020, S. 16282–16292.
- [47] Yongjie Shi, Xianghua Ying und Jinfa Yang. „Deep Unsupervised Domain Adaptation with Time Series Sensor Data: A Survey“. In: *Sensors* 22.15 (2022), S. 5507.
- [48] Allan Stisen u. a. „Smart Devices are Different: Assessing and Mitigating Mobile Sensing Heterogeneities for Activity Recognition“. In: *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems*. 2015, S. 127–140.
- [49] Baochen Sun und Kate Saenko. „Deep CORAL: Correlation Alignment for Deep Domain Adaptation“. In: *ECCV Workshops*. 2016.
- [50] Timo Sztyler und Heiner Stuckenschmidt. „On-body Localization of Wearable Devices: An Investigation of Position-Aware Activity Recognition“. In: *International Conference on Pervasive Computing and Communications*. 2016, S. 1–9.

- [51] Chi Ian Tang u. a. „SelfHAR: Improving Human Activity Recognition through Self-training with Unlabeled Data“. In: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5.1 (2021), S. 1–30.
- [52] Eric Tzeng u. a. „Adversarial Discriminative Domain Adaptation“. In: *IEEE Conference on Computer Vision and Pattern Recognition*. 2017, S. 7167–7176.
- [53] Qingchen Wang u. a. „TSFA: A Two-Stage Feature Alignment Method for Unsupervised Open-Set Domain Adaptation in Time-Series Classification“. In: *IEEE Transactions on Neural Networks and Learning Systems* (2026). Early access.
- [54] Wei Wang u. a. „COSDA: Counterfactual-Based Susceptibility Risk Framework for Open-Set Domain Adaptation“. In: *ICML*. 2025, S. 65845–65866.
- [55] Garrett Wilson, Janardhan Rao Doppa und Diane J. Cook. „CALDA: Improving Multi-Source Time Series Domain Adaptation with Contrastive Adversarial Learning“. In: *IEEE TPAMI* 45.12 (2023), S. 14208–14221.
- [56] Garrett Wilson, Janardhan Rao Doppa und Diane Joyce Cook. „Multi-Source Deep Domain Adaptation with Weak Supervision for Time-Series Sensor Data“. In: *ACM SIGKDD*. 2020, S. 1768–1778.
- [57] Yucheng Yang, Xiang Gu und Jian Sun. „Prototypical Partial Optimal Transport for Universal Domain Adaptation“. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2023.
- [58] Jui-Cheng Yen, Fu-Juay Chang und Shyang Chang. „A New Criterion for Automatic Multilevel Thresholding“. In: *IEEE Transactions on Image Processing* 4.3 (1995), S. 370–378.
- [59] Kaichao You u. a. „Universal Domain Adaptation“. In: *IEEE/CVF CVPR*. 2019, S. 2720–2729.
- [60] Huafeng Zhou u. a. „Bearing Fault Diagnosis Based on Partial Domain Adaptation Adversarial Network“. In: *Measurement Science and Technology* 33 (2022), S. 124003.

7.1 Extended Results

This appendix collects the full per-pair result tables for every scenario in our benchmark. The aggregated discussion in the main body uses these numbers as its source.

7.1.1 Closed-Set Domain Adaptation

All closed-set baselines and the UniDA methods run with the four shared known classes only (no n_{unknown} axis). Cells report target F1 (mean $\pm$ std across seeds).

Tabelle 7.1: closed set: Target F1 (mean $\pm$ std across seeds). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
ACON	0.903 $\pm$ 0.010	0.595 $\pm$ 0.089	0.869 $\pm$ 0.029	0.870 $\pm$ 0.145	0.911 $\pm$ 0.011	0.816 $\pm$ 0.170	0.827
AdvSKM	0.518 $\pm$ 0.028	0.416 $\pm$ 0.039	0.550 $\pm$ 0.011	0.171 $\pm$ 0.051	0.231 $\pm$ 0.035	0.233 $\pm$ 0.050	0.353
CDAN	0.865 $\pm$ 0.065	0.362 $\pm$ 0.145	0.851 $\pm$ 0.035	0.441 $\pm$ 0.133	0.300 $\pm$ 0.050	0.283 $\pm$ 0.115	0.517
CLUDA	0.815 $\pm$ 0.072	0.380 $\pm$ 0.095	0.780 $\pm$ 0.061	0.376 $\pm$ 0.096	0.414 $\pm$ 0.029	0.360 $\pm$ 0.185	0.521
CoDATS	0.797 $\pm$ 0.016	0.369 $\pm$ 0.109	0.779 $\pm$ 0.017	0.357 $\pm$ 0.156	0.418 $\pm$ 0.095	0.490 $\pm$ 0.212	0.535
CoTMix	0.807 $\pm$ 0.160	0.297 $\pm$ 0.144	0.766 $\pm$ 0.108	0.145 $\pm$ 0.051	0.322 $\pm$ 0.127	0.350 $\pm$ 0.176	0.448
DAAN	0.803 $\pm$ 0.013	0.326 $\pm$ 0.032	0.790 $\pm$ 0.010	0.278 $\pm$ 0.162	0.181 $\pm$ 0.147	0.148 $\pm$ 0.036	0.421
DANN	0.777 $\pm$ 0.089	0.382 $\pm$ 0.083	0.818 $\pm$ 0.066	0.327 $\pm$ 0.157	0.323 $\pm$ 0.052	0.521 $\pm$ 0.116	0.525
DDC	0.551 $\pm$ 0.056	0.414 $\pm$ 0.057	0.539 $\pm$ 0.026	0.103 $\pm$ 0.021	0.234 $\pm$ 0.022	0.227 $\pm$ 0.060	0.345
DIRT	0.872 $\pm$ 0.053	0.391 $\pm$ 0.032	0.875 $\pm$ 0.039	0.400 $\pm$ 0.063	0.333 $\pm$ 0.065	0.244 $\pm$ 0.079	0.519
DSAN	0.797 $\pm$ 0.035	0.440 $\pm$ 0.115	0.845 $\pm$ 0.040	0.414 $\pm$ 0.121	0.324 $\pm$ 0.033	0.198 $\pm$ 0.083	0.503
Deep_Coral	0.599 $\pm$ 0.044	0.369 $\pm$ 0.036	0.582 $\pm$ 0.025	0.248 $\pm$ 0.033	0.259 $\pm$ 0.040	0.204 $\pm$ 0.056	0.377
HoMM	0.637 $\pm$ 0.027	0.381 $\pm$ 0.067	0.619 $\pm$ 0.048	0.181 $\pm$ 0.072	0.275 $\pm$ 0.049	0.222 $\pm$ 0.059	0.386
MMDA	0.693 $\pm$ 0.072	0.287 $\pm$ 0.055	0.649 $\pm$ 0.038	0.356 $\pm$ 0.096	0.312 $\pm$ 0.031	0.362 $\pm$ 0.082	0.443
SASA	0.842 $\pm$ 0.015	0.381 $\pm$ 0.066	0.814 $\pm$ 0.031	0.448 $\pm$ 0.120	0.316 $\pm$ 0.058	0.291 $\pm$ 0.102	0.515
SSSS_TSA	0.926 $\pm$ 0.027	0.423 $\pm$ 0.017	0.841 $\pm$ 0.017	0.720 $\pm$ 0.015	0.477 $\pm$ 0.043	0.548 $\pm$ 0.128	0.656
SWL_Adapt	0.762 $\pm$ 0.089	0.311 $\pm$ 0.083	0.822 $\pm$ 0.020	0.319 $\pm$ 0.079	0.343 $\pm$ 0.069	0.335 $\pm$ 0.211	0.482
uDA	0.827 $\pm$ 0.070	0.359 $\pm$ 0.030	0.781 $\pm$ 0.012	0.386 $\pm$ 0.042	0.321 $\pm$ 0.019	0.186 $\pm$ 0.065	0.477
UDA	0.681 $\pm$ 0.030	0.539 $\pm$ 0.122	0.615 $\pm$ 0.040	0.212 $\pm$ 0.098	0.488 $\pm$ 0.153	0.560 $\pm$ 0.191	0.516
OVANet	0.725 $\pm$ 0.152	0.403 $\pm$ 0.114	0.681 $\pm$ 0.103	0.236 $\pm$ 0.139	0.305 $\pm$ 0.113	0.573 $\pm$ 0.097	0.487
DANCE	0.157 $\pm$ 0.022	0.087 $\pm$ 0.052	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.041
PPOT	0.770 $\pm$ 0.032	0.564 $\pm$ 0.102	0.002 $\pm$ 0.002	0.035 $\pm$ 0.064	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.228
UniOT	0.432 $\pm$ 0.153	0.446 $\pm$ 0.172	0.403 $\pm$ 0.041	0.267 $\pm$ 0.080	0.254 $\pm$ 0.047	0.190 $\pm$ 0.084	0.332
UniJDOT	0.620 $\pm$ 0.059	0.554 $\pm$ 0.243	0.589 $\pm$ 0.132	0.364 $\pm$ 0.073	0.222 $\pm$ 0.108	0.309 $\pm$ 0.122	0.443
RAINCOAT	0.424 $\pm$ 0.064	0.244 $\pm$ 0.064	0.455 $\pm$ 0.044	0.184 $\pm$ 0.061	0.347 $\pm$ 0.081	0.224 $\pm$ 0.184	0.313

7.1.2 Open-Set Domain Adaptation (OSDA)

OSDA-native methods (OSBP, TSFA) and the seven UniDA methods. Cells report mean $\pm$ std across seeds and over all n_{unknown} values evaluated for that pair.

Tabelle 7.2: OSDA: H-score (mean $\pm$ std across seeds and n). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
OSBP	0.700 $\pm$ 0.044	0.530 $\pm$ 0.147	0.001 $\pm$ 0.001	0.257 $\pm$ 0.199	0.298 $\pm$ 0.161	0.307 $\pm$ 0.068	0.349
TSFA	0.371 $\pm$ 0.037	0.341 $\pm$ 0.077	0.395 $\pm$ 0.057	0.290 $\pm$ 0.074	0.452 $\pm$ 0.072	0.318 $\pm$ 0.022	0.361
UDA	0.413 $\pm$ 0.024	0.403 $\pm$ 0.069	0.428 $\pm$ 0.027	0.337 $\pm$ 0.094	0.361 $\pm$ 0.067	0.367 $\pm$ 0.065	0.385
OVANet	0.259 $\pm$ 0.173	0.290 $\pm$ 0.174	0.118 $\pm$ 0.025	0.153 $\pm$ 0.099	0.291 $\pm$ 0.048	0.322 $\pm$ 0.079	0.239
DANCE	0.222 $\pm$ 0.116	0.114 $\pm$ 0.072	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.056
PPOT	0.172 $\pm$ 0.060	0.408 $\pm$ 0.103	0.014 $\pm$ 0.028	0.037 $\pm$ 0.081	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.105
UniOT	0.548 $\pm$ 0.124	0.480 $\pm$ 0.136	0.464 $\pm$ 0.052	0.259 $\pm$ 0.122	0.358 $\pm$ 0.109	0.318 $\pm$ 0.063	0.405
UniJDOT	0.563 $\pm$ 0.079	0.426 $\pm$ 0.163	0.462 $\pm$ 0.126	0.386 $\pm$ 0.177	0.373 $\pm$ 0.156	0.278 $\pm$ 0.088	0.415
RAINCOAT	0.222 $\pm$ 0.088	0.236 $\pm$ 0.100	0.314 $\pm$ 0.064	0.223 $\pm$ 0.097	0.171 $\pm$ 0.089	0.159 $\pm$ 0.045	0.221

Tabelle 7.3: OSDA: OS* (mean $\pm$ std across seeds and n). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
OSBP	0.644 $\pm$ 0.071	0.415 $\pm$ 0.153	0.710 $\pm$ 0.067	0.474 $\pm$ 0.188	0.630 $\pm$ 0.130	0.531 $\pm$ 0.156	0.568
TSFA	0.675 $\pm$ 0.135	0.564 $\pm$ 0.159	0.574 $\pm$ 0.045	0.425 $\pm$ 0.153	0.716 $\pm$ 0.053	0.437 $\pm$ 0.056	0.565
UDA	0.554 $\pm$ 0.050	0.556 $\pm$ 0.137	0.508 $\pm$ 0.039	0.334 $\pm$ 0.121	0.349 $\pm$ 0.102	0.372 $\pm$ 0.117	0.445
OVANet	0.604 $\pm$ 0.129	0.494 $\pm$ 0.134	0.771 $\pm$ 0.059	0.322 $\pm$ 0.094	0.340 $\pm$ 0.117	0.551 $\pm$ 0.154	0.514
DANCE	0.138 $\pm$ 0.078	0.069 $\pm$ 0.051	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.035
PPOT	0.846 $\pm$ 0.035	0.512 $\pm$ 0.081	0.007 $\pm$ 0.015	0.021 $\pm$ 0.047	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.231
UniOT	0.445 $\pm$ 0.111	0.370 $\pm$ 0.119	0.347 $\pm$ 0.051	0.163 $\pm$ 0.089	0.241 $\pm$ 0.083	0.218 $\pm$ 0.063	0.298
UniJDOT	0.598 $\pm$ 0.099	0.422 $\pm$ 0.157	0.415 $\pm$ 0.104	0.307 $\pm$ 0.161	0.328 $\pm$ 0.168	0.249 $\pm$ 0.110	0.387
RAINCOAT	0.283 $\pm$ 0.065	0.300 $\pm$ 0.125	0.406 $\pm$ 0.065	0.236 $\pm$ 0.095	0.279 $\pm$ 0.164	0.131 $\pm$ 0.027	0.273

Tabelle 7.4: OSDA: UNK (mean $\pm$ std across seeds and n). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
OSBP	0.772 $\pm$ 0.048	0.793 $\pm$ 0.128	0.001 $\pm$ 0.001	0.243 $\pm$ 0.191	0.238 $\pm$ 0.170	0.225 $\pm$ 0.057	0.379
TSFA	0.259 $\pm$ 0.023	0.255 $\pm$ 0.072	0.303 $\pm$ 0.060	0.235 $\pm$ 0.055	0.334 $\pm$ 0.069	0.252 $\pm$ 0.020	0.273
UDA	0.333 $\pm$ 0.036	0.328 $\pm$ 0.060	0.372 $\pm$ 0.037	0.359 $\pm$ 0.083	0.394 $\pm$ 0.042	0.382 $\pm$ 0.027	0.362
OVANet	0.192 $\pm$ 0.154	0.327 $\pm$ 0.290	0.064 $\pm$ 0.015	0.129 $\pm$ 0.098	0.319 $\pm$ 0.143	0.250 $\pm$ 0.091	0.213
DANCE	0.873 $\pm$ 0.136	0.910 $\pm$ 0.154	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.964
PPOT	0.097 $\pm$ 0.039	0.355 $\pm$ 0.116	0.997 $\pm$ 0.010	0.987 $\pm$ 0.035	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.739
UniOT	0.756 $\pm$ 0.122	0.755 $\pm$ 0.154	0.708 $\pm$ 0.061	0.780 $\pm$ 0.095	0.740 $\pm$ 0.168	0.647 $\pm$ 0.081	0.731
UniJDOT	0.547 $\pm$ 0.100	0.561 $\pm$ 0.262	0.649 $\pm$ 0.266	0.662 $\pm$ 0.198	0.571 $\pm$ 0.263	0.383 $\pm$ 0.127	0.562
RAINCOAT	0.215 $\pm$ 0.101	0.288 $\pm$ 0.168	0.275 $\pm$ 0.109	0.292 $\pm$ 0.147	0.282 $\pm$ 0.266	0.250 $\pm$ 0.129	0.267

7.1.3 Partial Domain Adaptation (PDA)

PDA-native methods (SPADA, PDAAN) and the seven UniDA methods. Cells report target F1 (mean $\pm$ std across seeds and n_{unknown}).

Tabelle 7.5: PDA: Target F1 (mean $\pm$ std across seeds and n). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
SPADA	0.757 $\pm$ 0.081	0.227 $\pm$ 0.152	0.303 $\pm$ 0.081	0.267 $\pm$ 0.080	0.459 $\pm$ 0.134	0.465 $\pm$ 0.166	0.413
PDAAN	0.742 $\pm$ 0.085	0.405 $\pm$ 0.224	0.032 $\pm$ 0.035	0.021 $\pm$ 0.033	0.345 $\pm$ 0.134	0.384 $\pm$ 0.194	0.321
UDA	0.574 $\pm$ 0.062	0.386 $\pm$ 0.131	0.217 $\pm$ 0.054	0.175 $\pm$ 0.090	0.296 $\pm$ 0.099	0.270 $\pm$ 0.140	0.320
OVANet	0.245 $\pm$ 0.203	0.143 $\pm$ 0.150	0.005 $\pm$ 0.009	0.001 $\pm$ 0.007	0.101 $\pm$ 0.115	0.074 $\pm$ 0.103	0.095
DANCE	0.055 $\pm$ 0.060	0.033 $\pm$ 0.047	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.015
PPOT	0.456 $\pm$ 0.108	0.252 $\pm$ 0.136	0.000 $\pm$ 0.000	0.031 $\pm$ 0.063	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.123
UniOT	0.380 $\pm$ 0.227	0.237 $\pm$ 0.199	0.026 $\pm$ 0.038	0.018 $\pm$ 0.025	0.081 $\pm$ 0.117	0.106 $\pm$ 0.128	0.141
UniJDOT	0.549 $\pm$ 0.095	0.447 $\pm$ 0.186	0.435 $\pm$ 0.148	0.268 $\pm$ 0.179	0.163 $\pm$ 0.120	0.196 $\pm$ 0.088	0.343
RAINCOAT	0.378 $\pm$ 0.127	0.314 $\pm$ 0.120	0.147 $\pm$ 0.047	0.115 $\pm$ 0.086	0.210 $\pm$ 0.076	0.215 $\pm$ 0.158	0.230

7.1.4 Universal Domain Adaptation (UniDA)

The seven UniDA methods; both source-private classes are added to the source side AND target-private classes to the target side as n_{unknown} grows. Cells report mean $\pm$ std across seeds and n_{unknown} .

Tabelle 7.6: UniDA: H-score (mean $\pm$ std across seeds and n). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
UDA	0.434 $\pm$ 0.023	0.370 $\pm$ 0.068	0.285 $\pm$ 0.026	0.207 $\pm$ 0.066	0.320 $\pm$ 0.074	0.286 $\pm$ 0.100	0.317
OVANet	0.224 $\pm$ 0.162	0.095 $\pm$ 0.146	0.009 $\pm$ 0.008	0.000 $\pm$ 0.000	0.214 $\pm$ 0.109	0.084 $\pm$ 0.095	0.105
DANCE	0.086 $\pm$ 0.100	0.019 $\pm$ 0.032	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018
PPOT	0.480 $\pm$ 0.079	0.226 $\pm$ 0.090	0.000 $\pm$ 0.000	0.024 $\pm$ 0.083	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.122
UniOT	0.363 $\pm$ 0.285	0.330 $\pm$ 0.232	0.067 $\pm$ 0.049	0.014 $\pm$ 0.025	0.254 $\pm$ 0.137	0.108 $\pm$ 0.149	0.189
UniJDOT	0.594 $\pm$ 0.084	0.446 $\pm$ 0.165	0.502 $\pm$ 0.093	0.175 $\pm$ 0.159	0.317 $\pm$ 0.129	0.262 $\pm$ 0.155	0.382
RAINCOAT	0.205 $\pm$ 0.072	0.195 $\pm$ 0.080	0.079 $\pm$ 0.055	0.144 $\pm$ 0.084	0.115 $\pm$ 0.079	0.079 $\pm$ 0.035	0.136

Tabelle 7.7: UniDA: OS* (mean $\pm$ std across seeds and n). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
UDA	0.549 $\pm$ 0.060	0.407 $\pm$ 0.113	0.214 $\pm$ 0.028	0.159 $\pm$ 0.078	0.278 $\pm$ 0.102	0.244 $\pm$ 0.121	0.309
OVANet	0.300 $\pm$ 0.222	0.130 $\pm$ 0.182	0.008 $\pm$ 0.009	0.000 $\pm$ 0.000	0.156 $\pm$ 0.098	0.071 $\pm$ 0.104	0.111
DANCE	0.048 $\pm$ 0.060	0.010 $\pm$ 0.017	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.010
PPOT	0.418 $\pm$ 0.106	0.241 $\pm$ 0.108	0.000 $\pm$ 0.000	0.015 $\pm$ 0.052	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.112
UniOT	0.287 $\pm$ 0.237	0.238 $\pm$ 0.180	0.041 $\pm$ 0.033	0.007 $\pm$ 0.014	0.164 $\pm$ 0.093	0.071 $\pm$ 0.102	0.135
UniJDOT	0.570 $\pm$ 0.124	0.356 $\pm$ 0.174	0.485 $\pm$ 0.085	0.131 $\pm$ 0.149	0.216 $\pm$ 0.118	0.180 $\pm$ 0.131	0.323
RAINCOAT	0.228 $\pm$ 0.090	0.266 $\pm$ 0.111	0.122 $\pm$ 0.036	0.147 $\pm$ 0.084	0.163 $\pm$ 0.105	0.073 $\pm$ 0.072	0.166

Tabelle 7.8: UniDA: UNK (mean $\pm$ std across seeds and n). Bold = best per pair.

Algorithm	RW $\rightarrow$ PA	RW $\rightarrow$ MH	PA $\rightarrow$ RW	PA $\rightarrow$ MH	MH $\rightarrow$ RW	MH $\rightarrow$ PA	Avg
UDA	0.362 ± 0.031	0.351 ± 0.056	0.434 ± 0.032	0.352 ± 0.054	0.421 ± 0.043	0.409 ± 0.026	0.388
OVANet	0.568 ± 0.375	0.726 ± 0.379	0.099 ± 0.148	0.381 ± 0.307	0.657 ± 0.243	0.680 ± 0.268	0.519
DANCE	0.622 ± 0.469	0.866 ± 0.299	0.994 ± 0.006	0.999 ± 0.002	1.000 ± 0.000	1.000 ± 0.000	0.914
PPOT	0.595 ± 0.108	0.295 ± 0.167	0.887 ± 0.105	0.797 ± 0.109	1.000 ± 0.000	1.000 ± 0.000	0.762
UniOT	0.869 ± 0.108	0.870 ± 0.131	0.333 ± 0.133	0.641 ± 0.222	0.756 ± 0.155	0.866 ± 0.173	0.723
UniJDOT	0.675 ± 0.156	0.760 ± 0.168	0.548 ± 0.170	0.561 ± 0.187	0.836 ± 0.174	0.686 ± 0.094	0.677
RAINCOAT	0.216 ± 0.086	0.218 ± 0.113	0.093 ± 0.099	0.215 ± 0.130	0.179 ± 0.206	0.183 ± 0.095	0.184